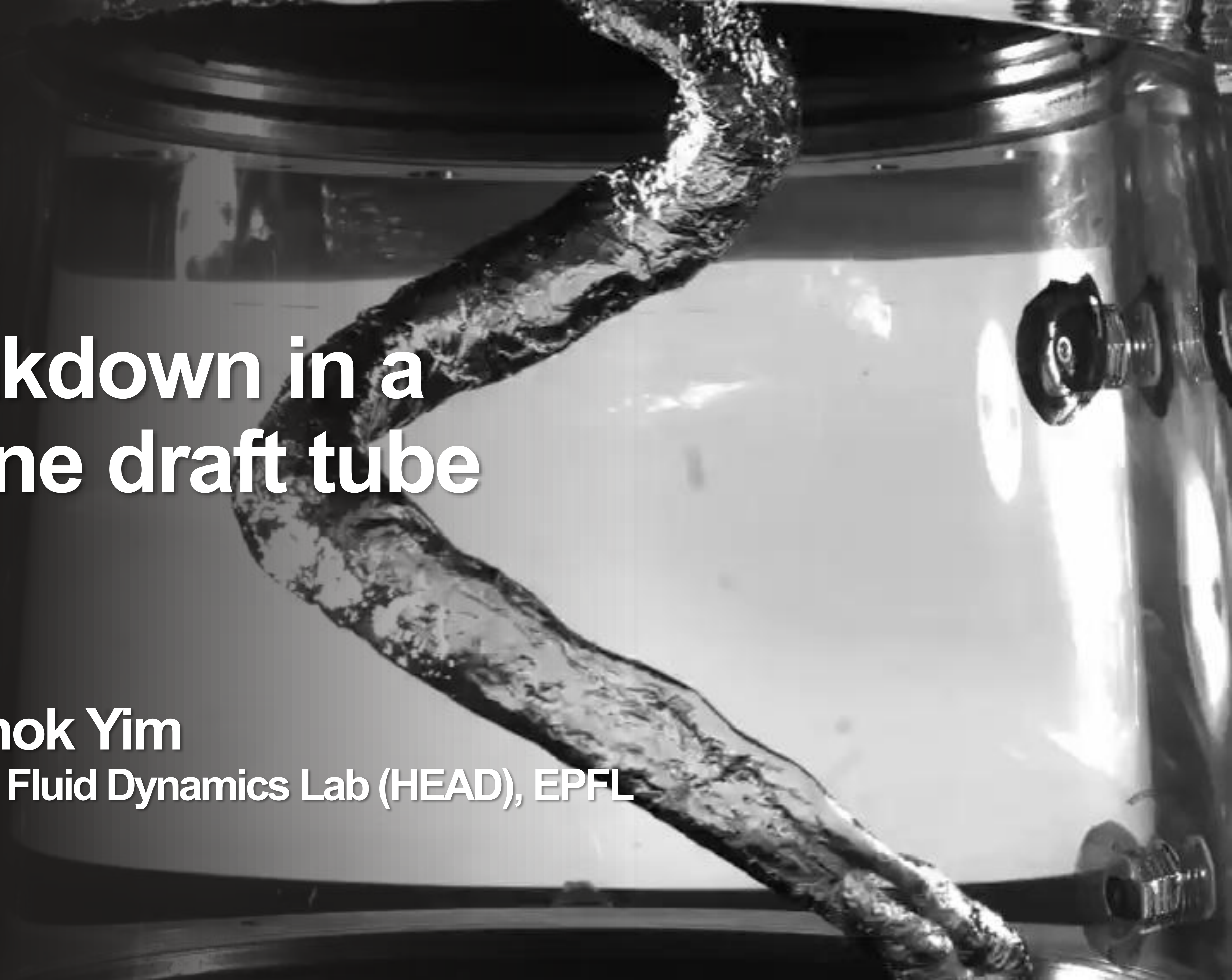
EPFL

BIFD 2026

Vortex breakdown in a hydro turbine draft tube swirling jet

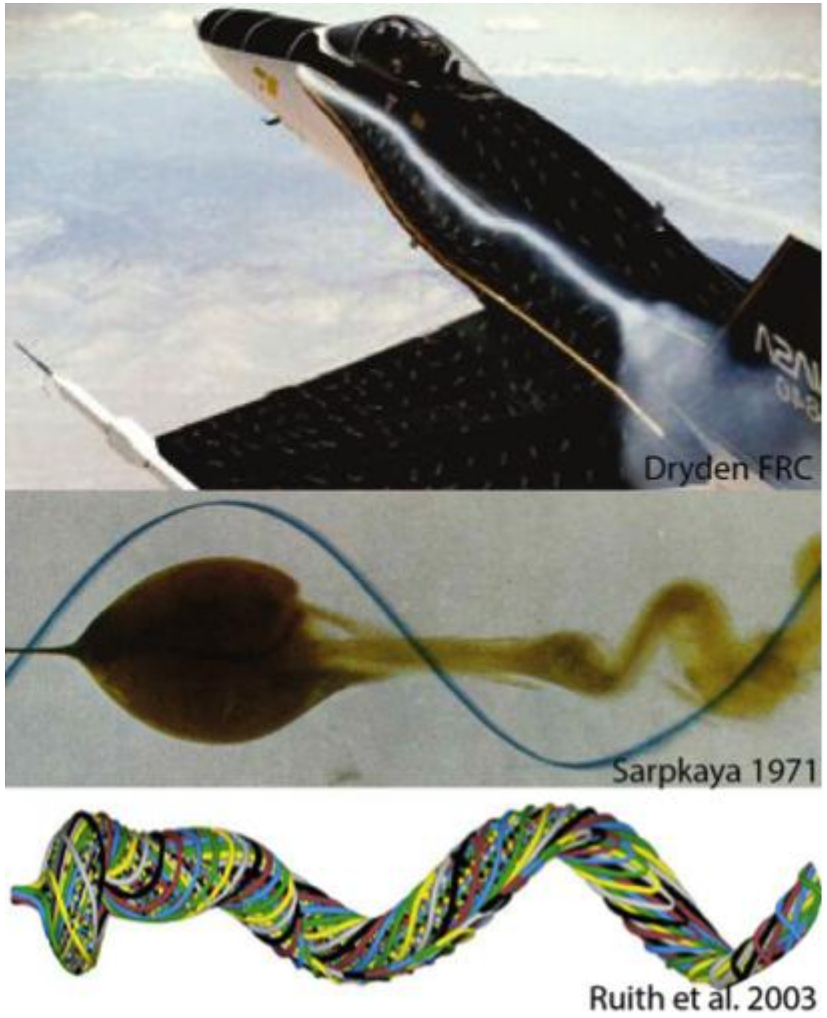


Artur Gesla & Eunok Yim

Hydro Energy & Applied Fluid Dynamics Lab (HEAD), EPFL

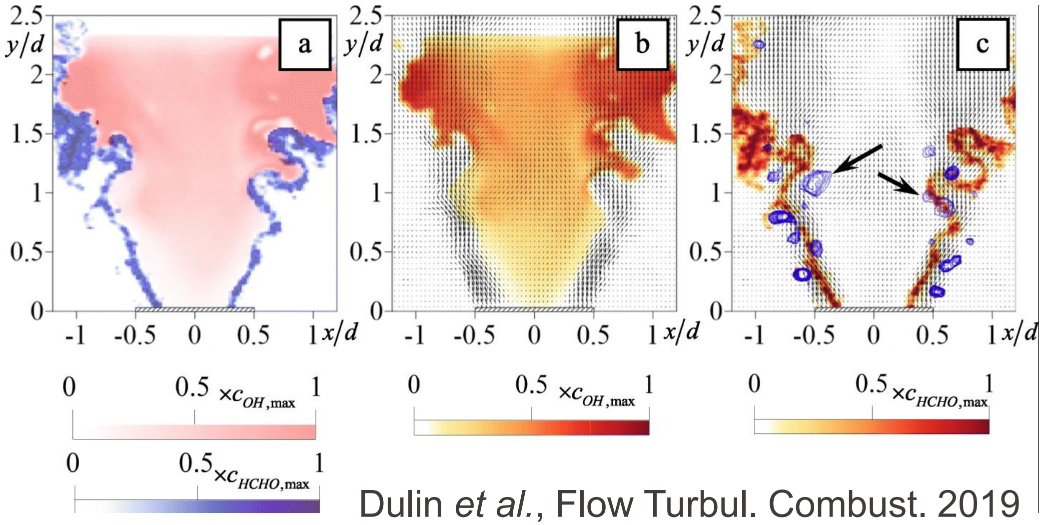
École polytechnique fédérale de Lausanne

Vortex breakdown



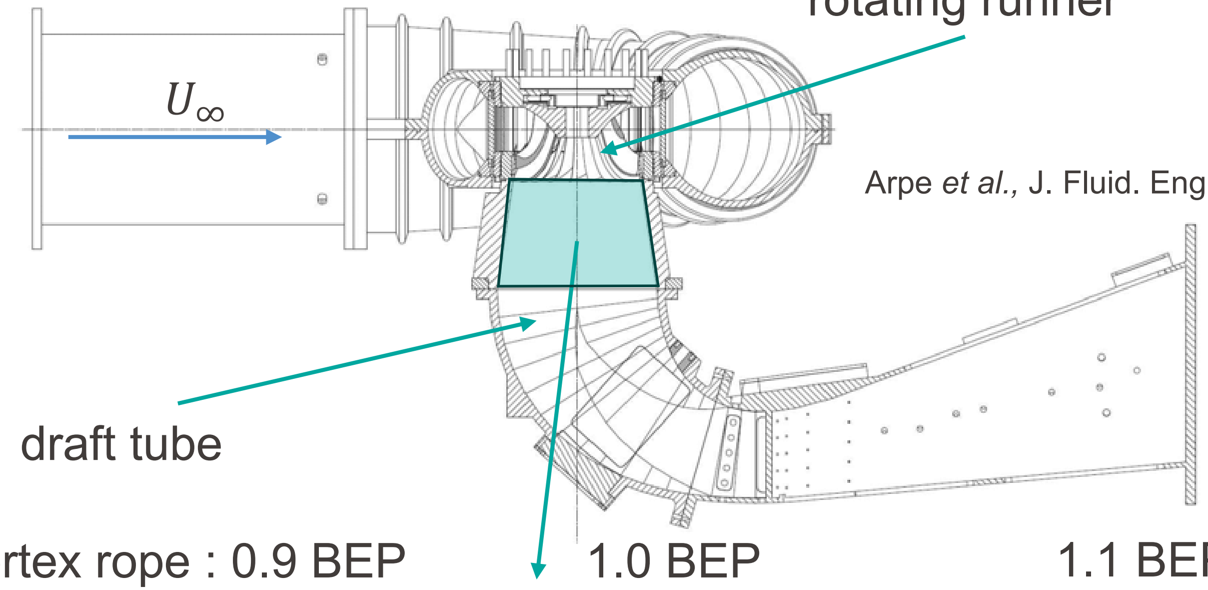
Viola, 2016

Pasche *et al.*, Exp. Fuids 2014



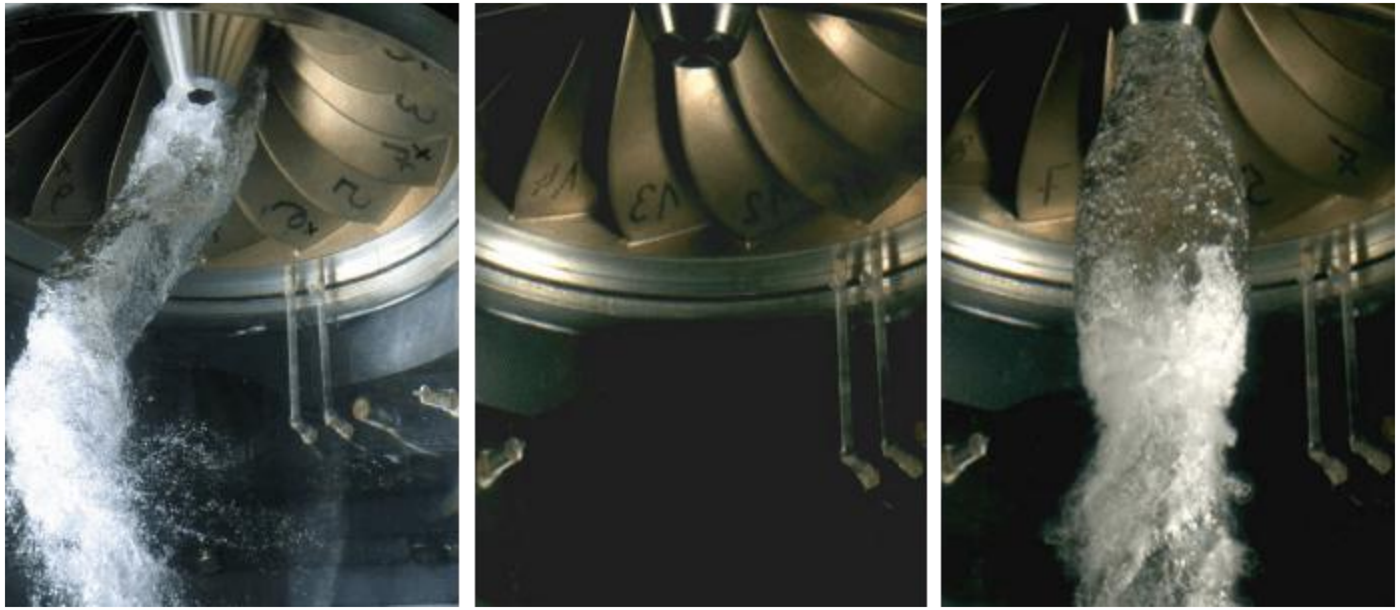
Dulin *et al.*, Flow Turbul. Combust. 2019

Francis turbine



Arpe *et al.*, J. Fluid. Eng. 2009

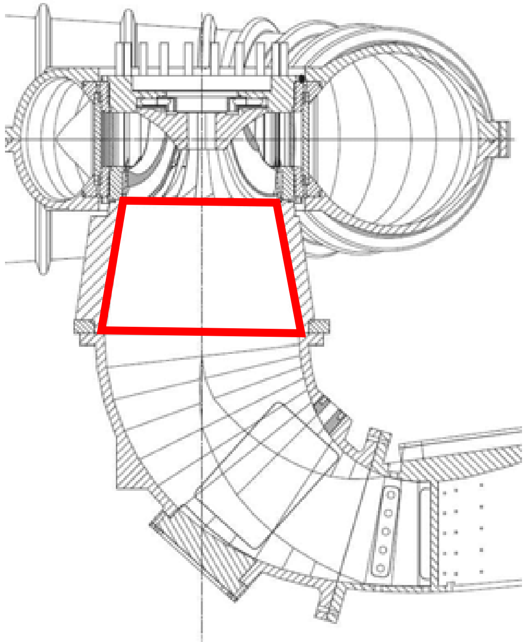
Vortex rope : 0.9 BEP 1.0 BEP 1.1 BEP

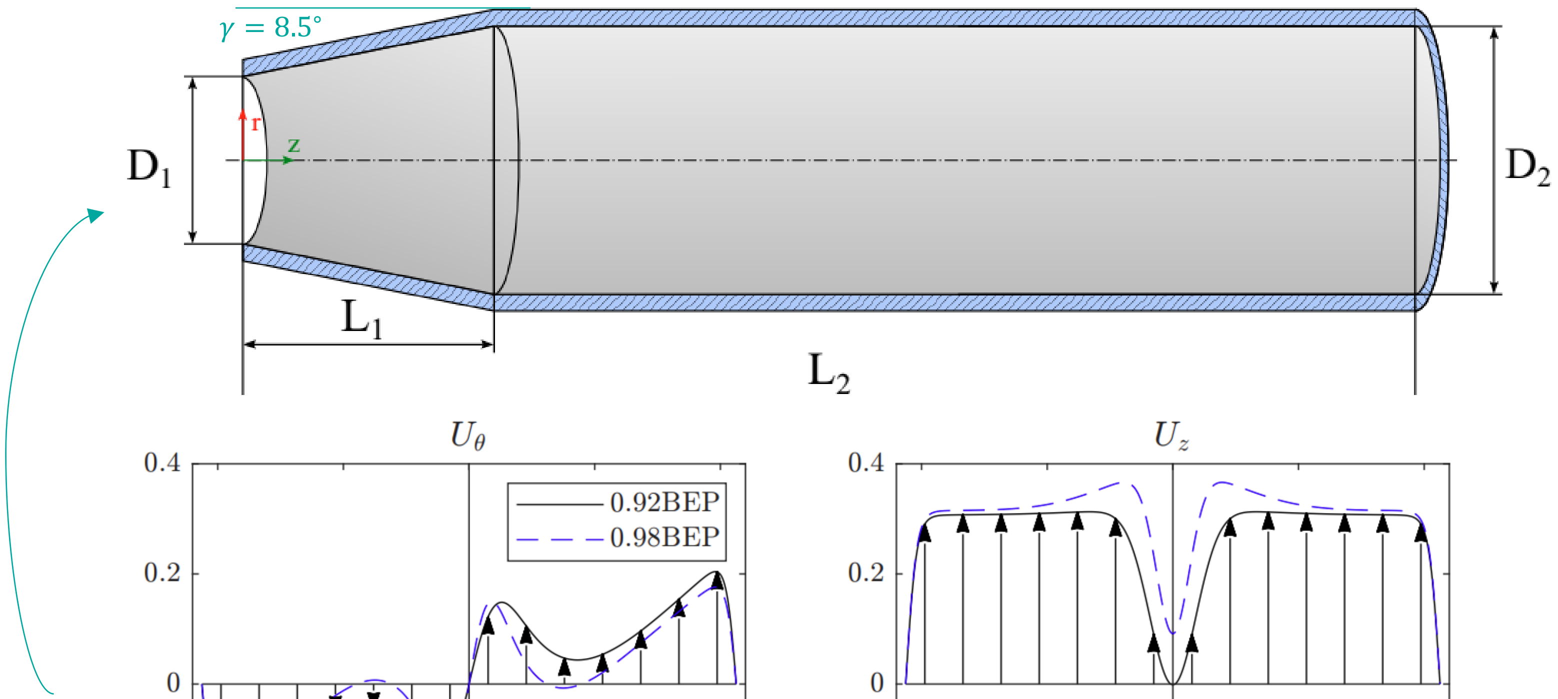


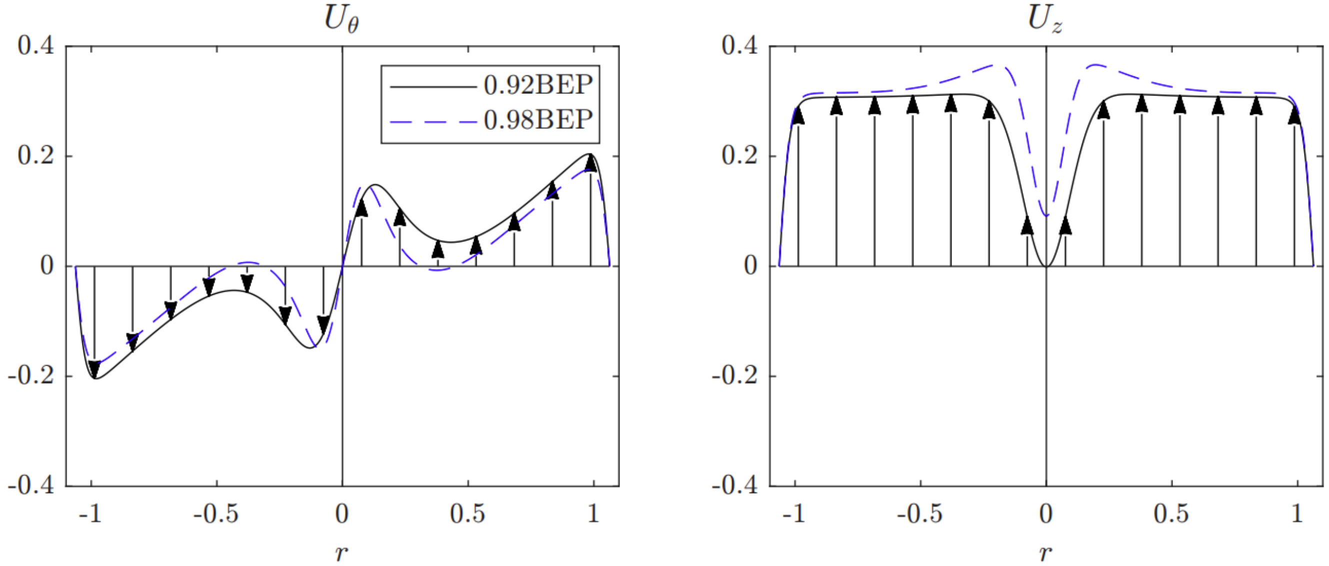
Seidel *et al.*, IOP Conf. Ser. 2014

Almini *et al.*, Exp. Fluids 2023

BEP – best operating point







$$Re = \frac{\Omega R^2}{\nu}$$

Ω - runner angular frequency
R – runner radius
ν – kinematic viscosity

Velocity profile fit to experimental data (Susan-Resiga *et al.*, JFE 2006).

$$U_r = 0,$$
$$U_\theta = \Omega_0 r + \frac{\Omega_1 r_1^2}{r}\left[1 - \exp(-r^2/r_1^2)\right] + \frac{\Omega_2 r_2^2}{r}\left[1 - \exp(-r^2/r_2^2)\right],$$
$$U_z = U_0 + U_1 \exp(-r^2/r_1^2) + U_2 \exp(-r^2/r_2^2).$$

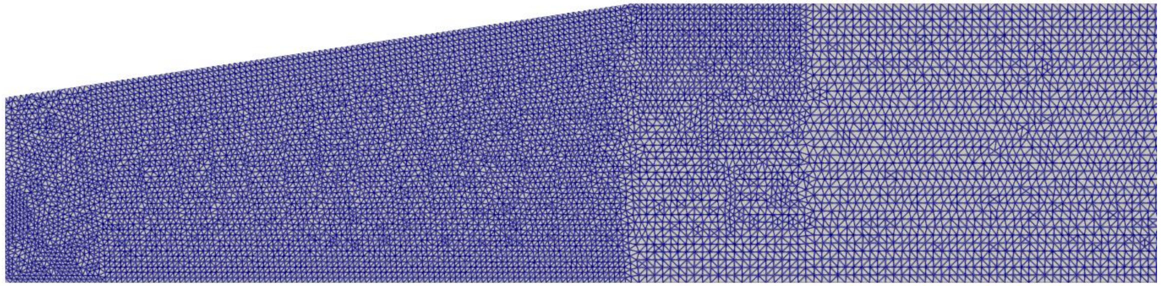
Methods

- Incompressible Navier-Stokes equation

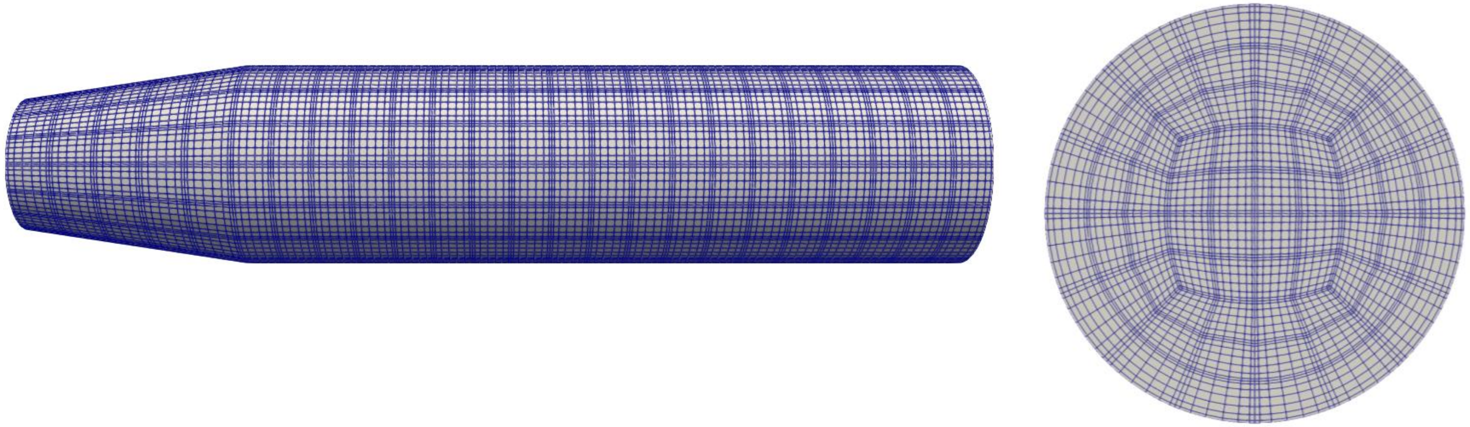
$$\begin{aligned} \frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \boldsymbol{\nabla})\boldsymbol{u} &= -\boldsymbol{\nabla} p + \frac{1}{Re}\boldsymbol{\nabla}^2 \boldsymbol{u}, \\ \boldsymbol{\nabla} \cdot \boldsymbol{u} &= 0. \end{aligned}$$

- Boundary conditions:
  - Inlet – Dirichlet, $\boldsymbol{u} = \boldsymbol{u}_{inlet}$,
  - Wall :
    - No-slip : $\boldsymbol{u} = 0$,
    - Free slip :
      - $(\boldsymbol{\tau} \cdot \boldsymbol{n}) \cdot \boldsymbol{t} = 0$,
      - $\boldsymbol{u} \cdot \boldsymbol{n} = 0$,
  - Outlet : $(Re^{-1}\boldsymbol{\nabla}\boldsymbol{u} - p\boldsymbol{I}) \cdot \boldsymbol{n} = 0$.

- FreeFem++ (finite elements, steady state)

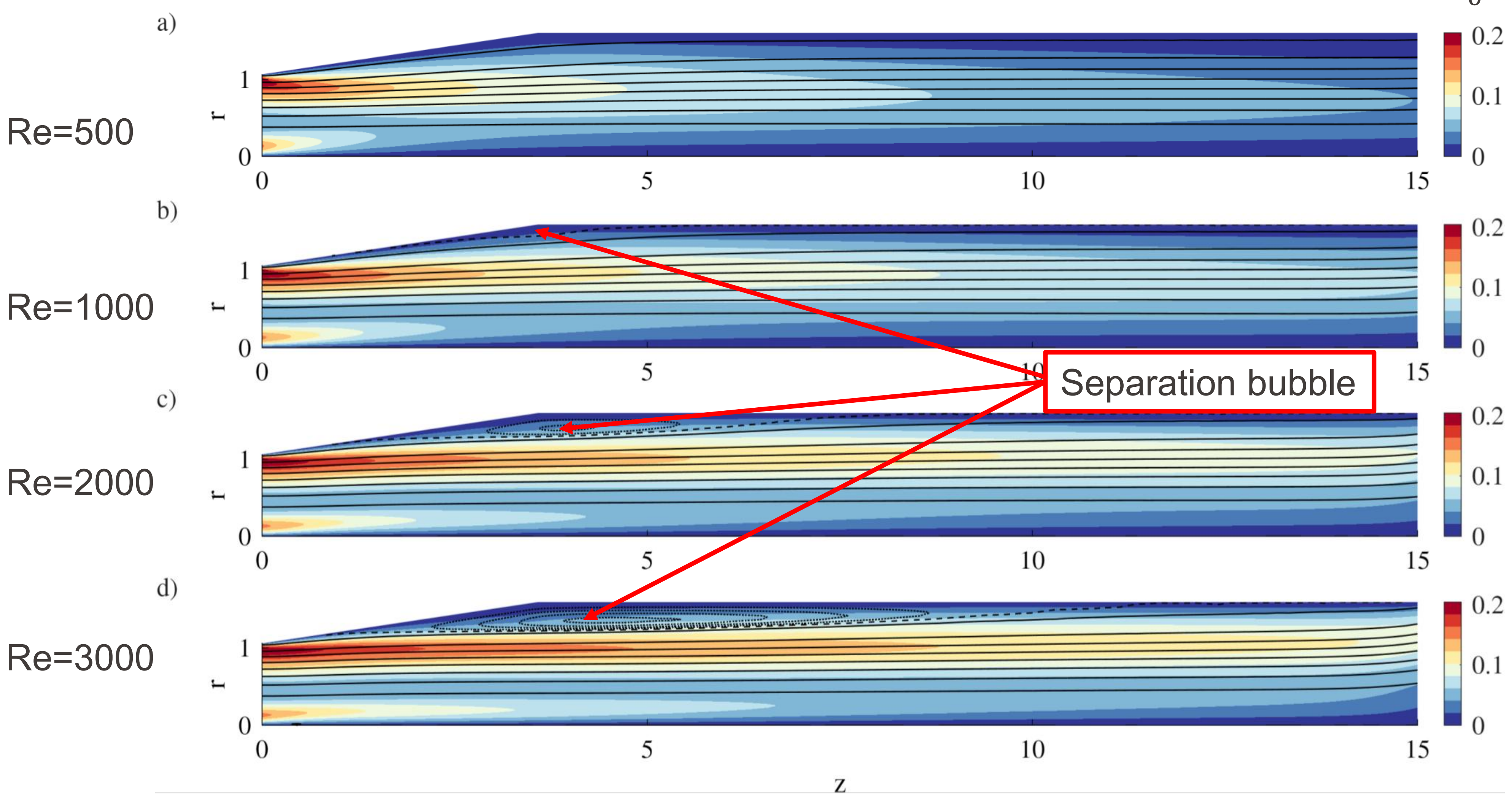


- nek5000 (spectral elements, time integration)



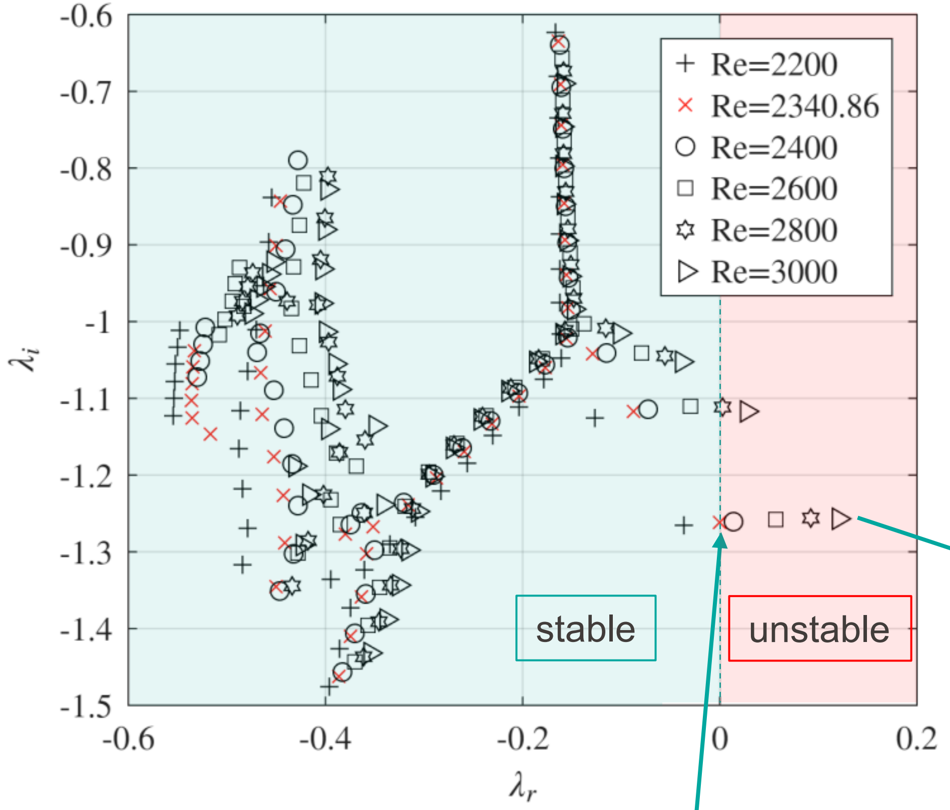
Base flow – no-slip wall (FreeFem++)

$\psi_{axis} = 0,\ \psi_{wall} = \int_0^R u_z r\, dr$



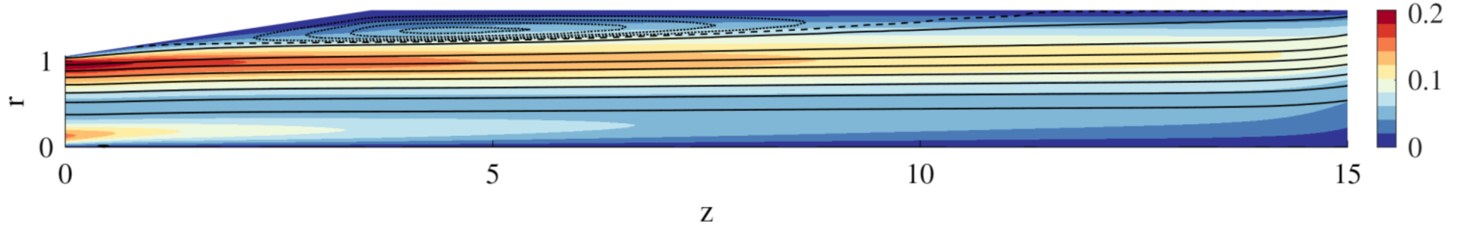
Linear stability

m=1



$Re_c = 2340.86$

Base flow Re=3000 :



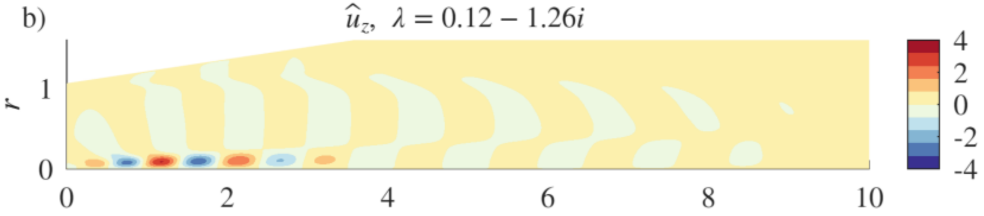
Perturbation : $\mathbf{u}(r,\theta,z,t) = \hat{\mathbf{u}}(r,z)\exp(im\theta + \lambda t)$

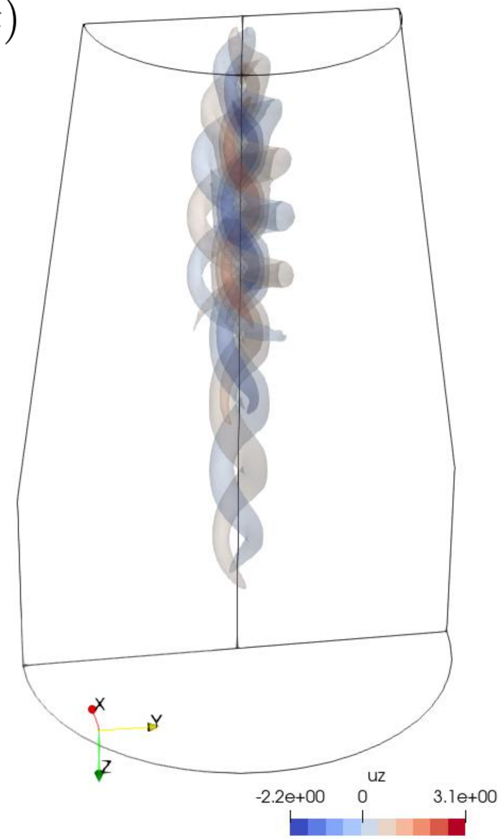
$$
\begin{aligned}
\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{U}_b \cdot \nabla)\mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{U}_b &= -\nabla p + \frac{1}{Re}\nabla^2 \mathbf{u} \\
\nabla \cdot \mathbf{u} &= 0
\end{aligned}
$$

$$\mathbf{q} = [\mathbf{u}, p]$$

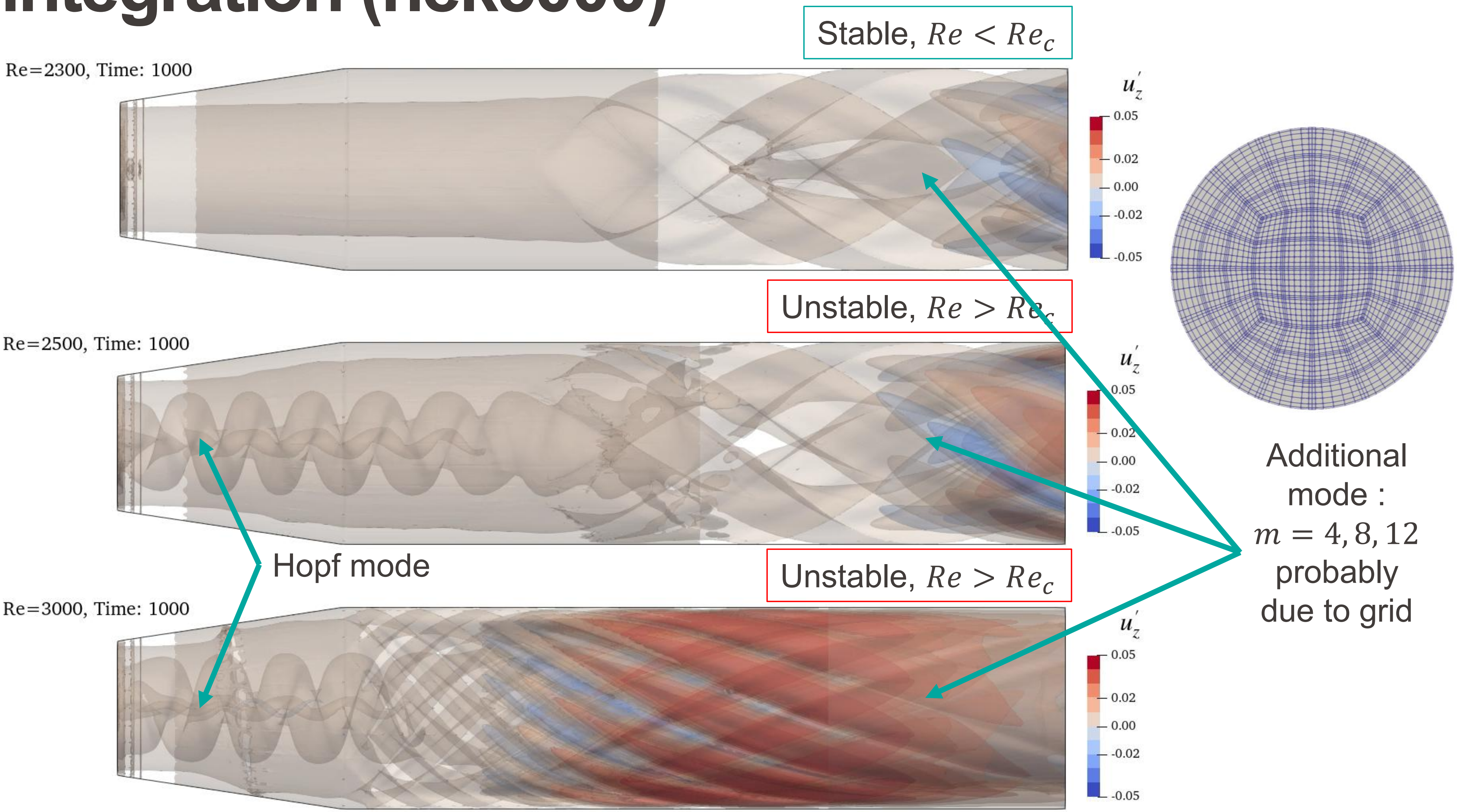
$$\mathbf{A}\mathbf{q} = \lambda \mathbf{B}\mathbf{q}$$

u_z eigenvector:



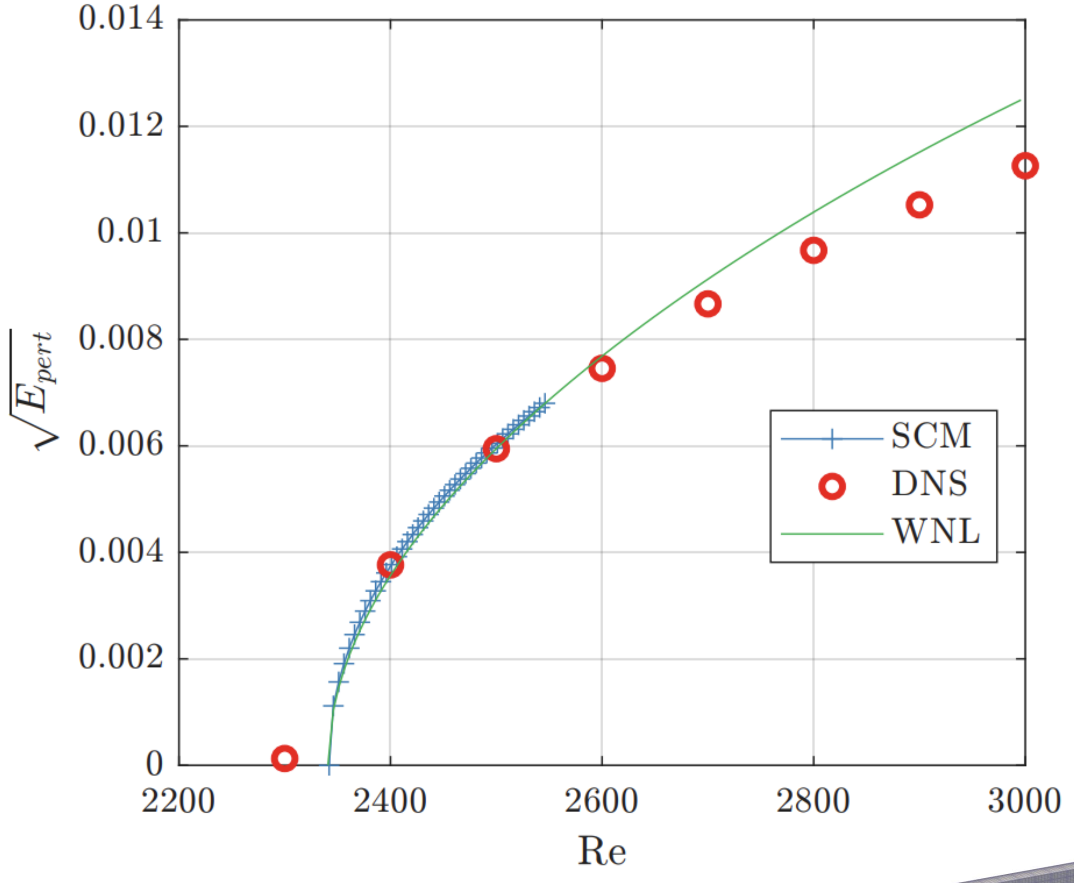


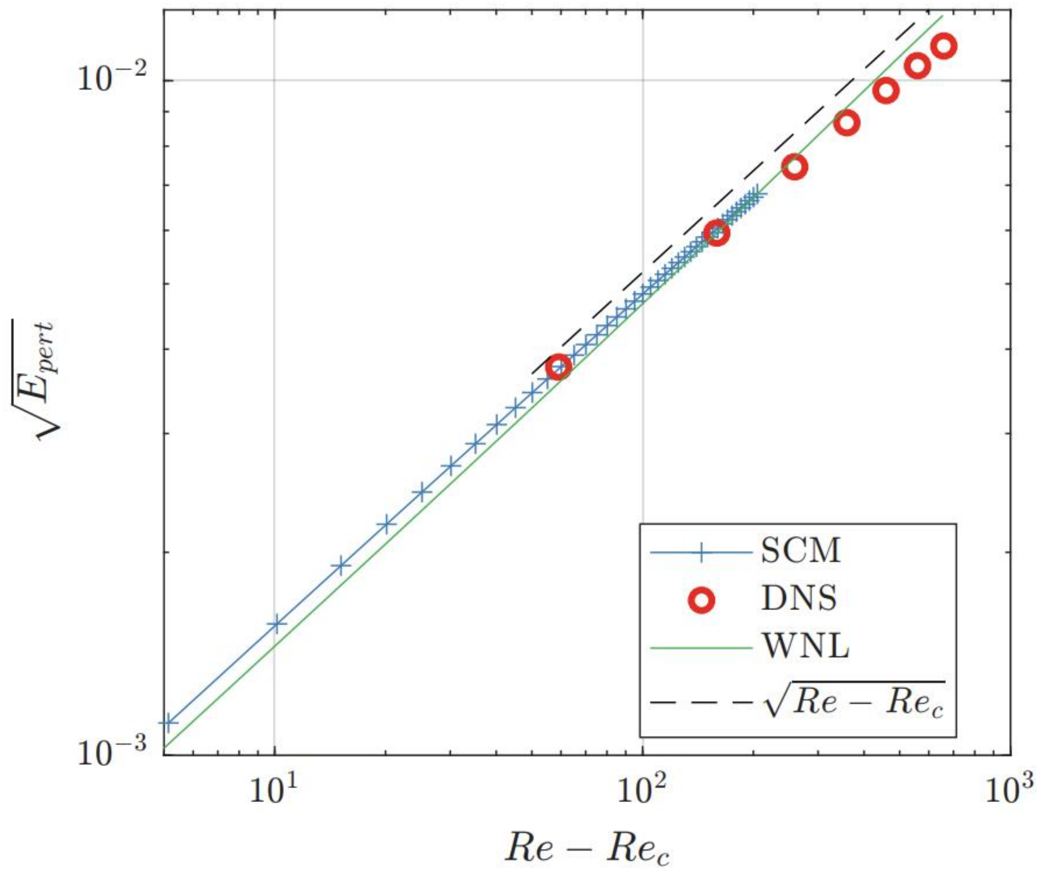
Time integration (nek5000)



Saturation level

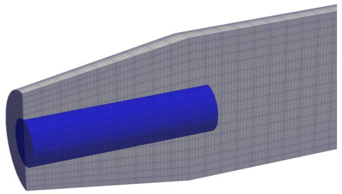
WNL – Weakly nonlinear analysis (e.g. Sipp & Lebedev, JFM 2007)
SCM – Self-Consistent Model (Mantič-Lugo *et al.*, PRL 2014)





$$E_{pert}(t) = \frac{1}{2}\int_0^4 \int_0^{2\pi} \int_0^{0.5} u_r'^2 + u_\theta'^2 + u_z'^2 \, r \,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}z,$$

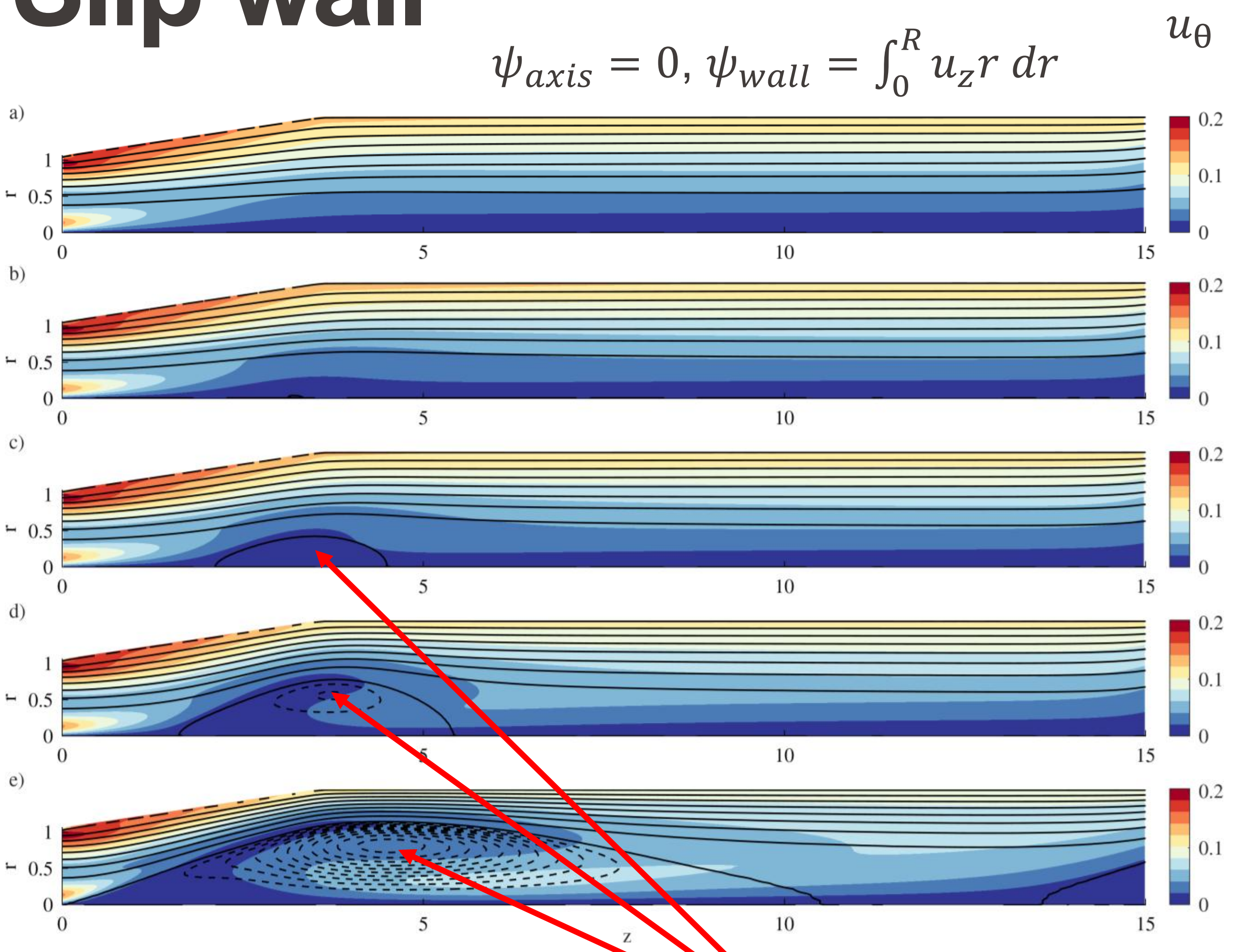
$$\boldsymbol{u}'(t) = \boldsymbol{u}(t) - \boldsymbol{U}_b$$



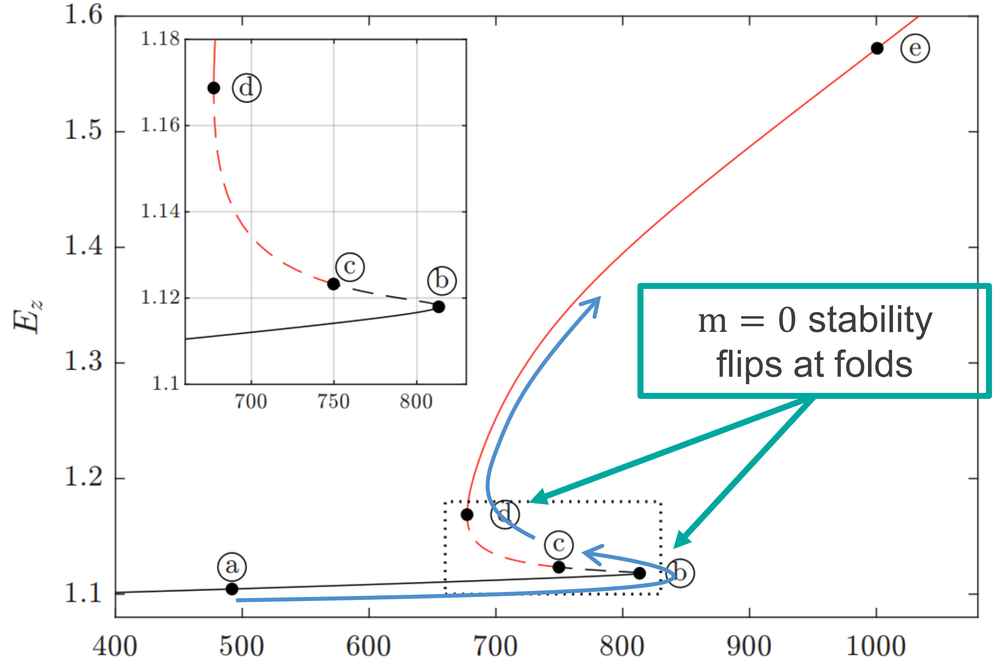
Supercritical Hopf bifurcation

Slip wall

$$\psi_{axis} = 0, \psi_{wall} = \int_0^R u_z r \, dr$$



Separation bubble, now at the axis

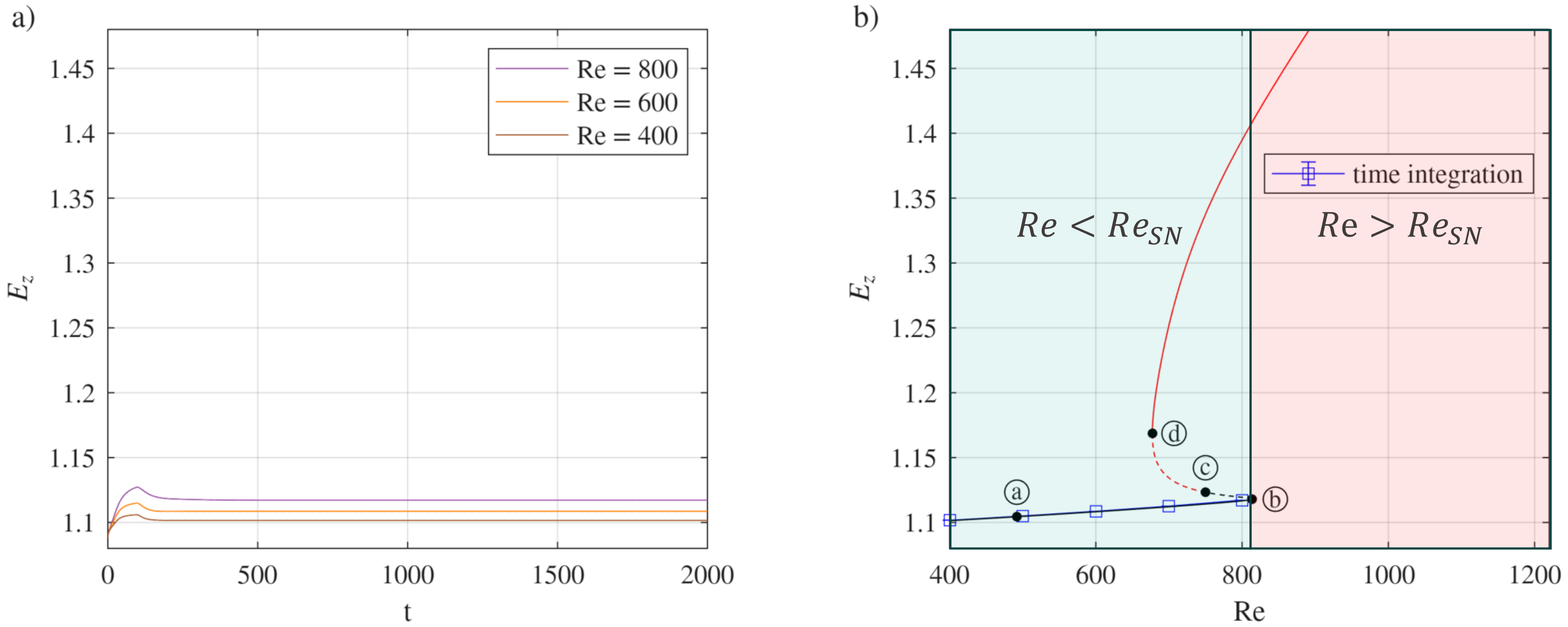


$$E_z = \frac{1}{2} \int_0^{15} \int_0^{2\pi} \int_0^{R(z)} u_z^2 \, r \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}z$$

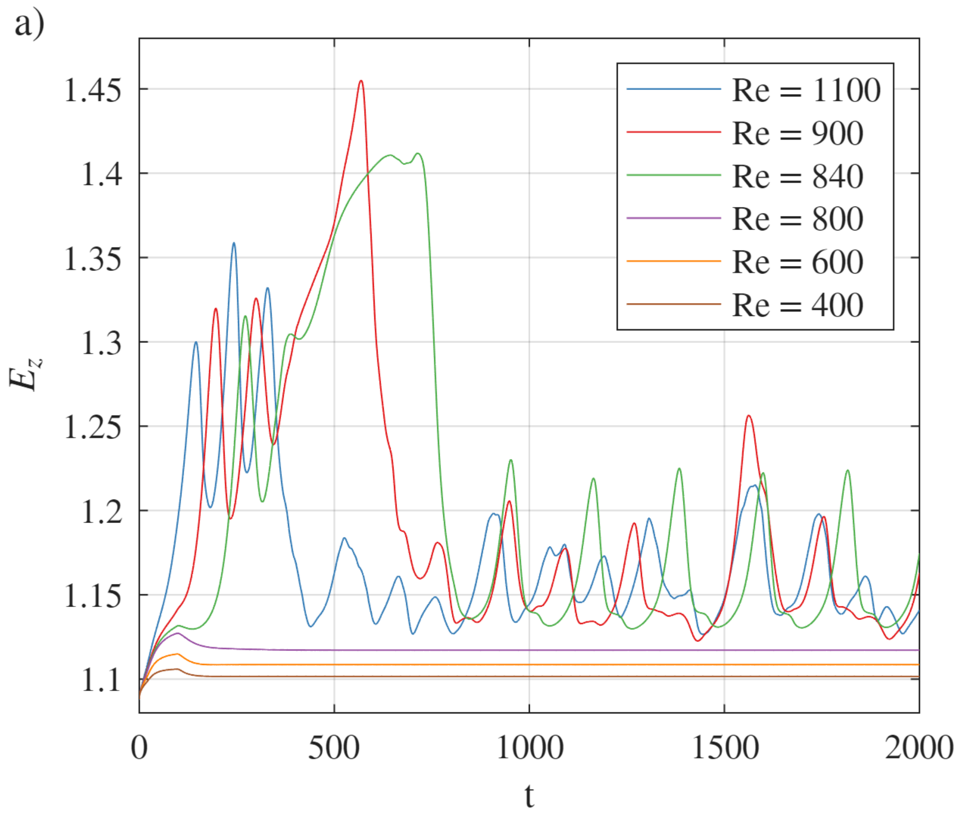
m = 0 stable / unstable

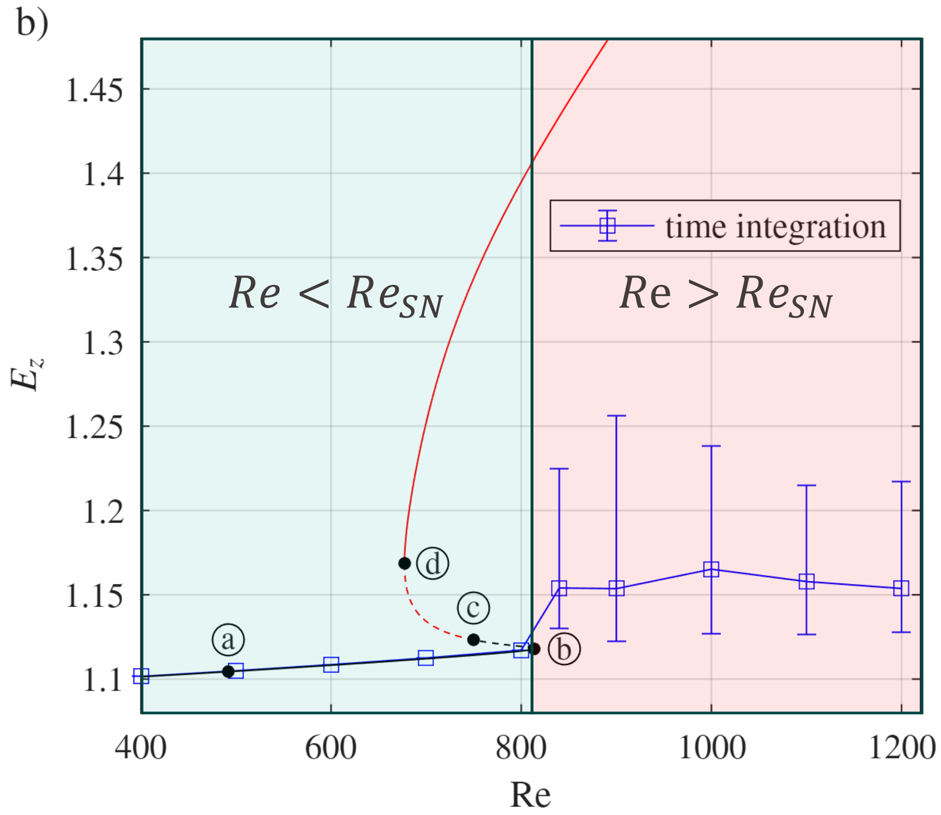
m = 1 stable / unstable

Time integration - slip wall

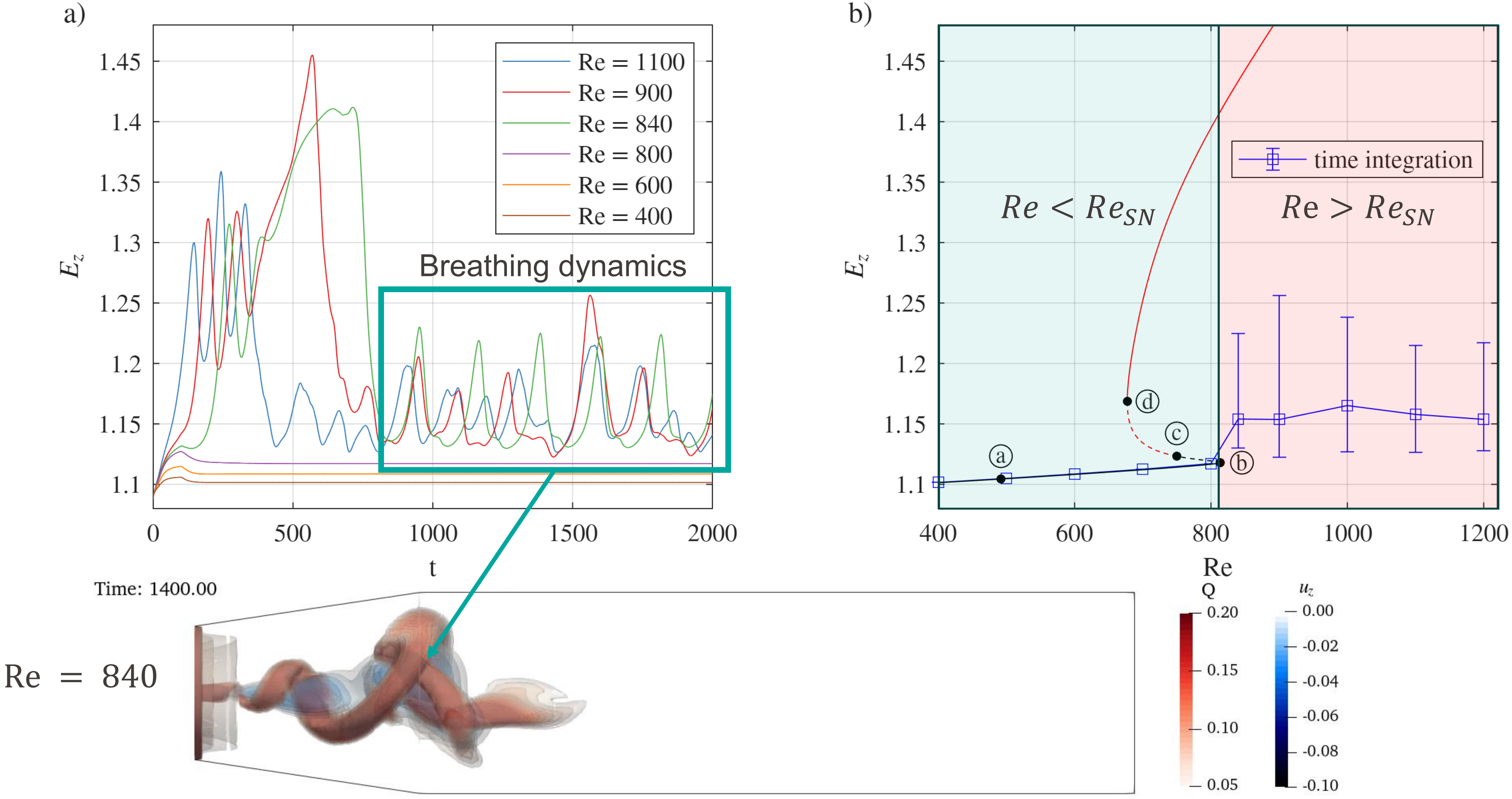


Time integration - slip wall



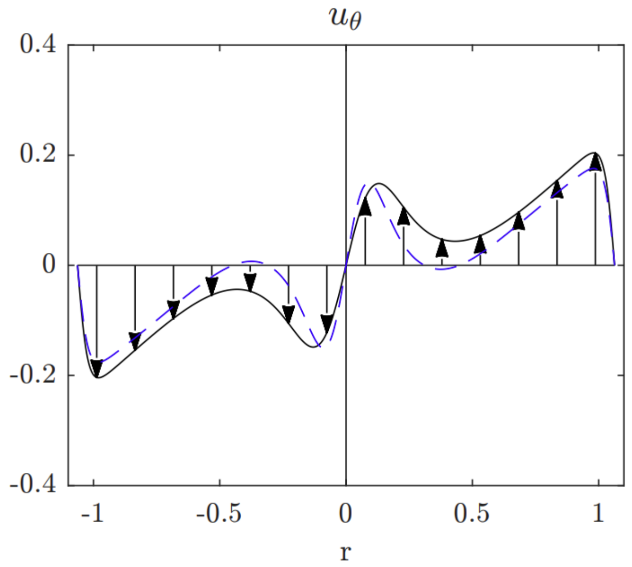


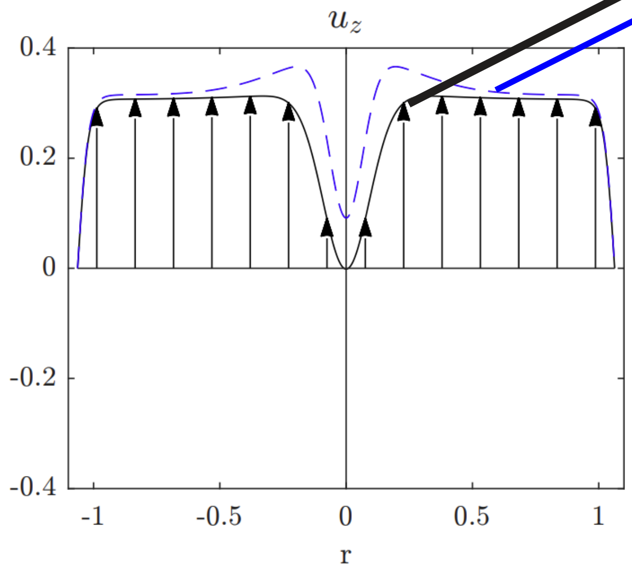
Time integration - slip wall



Operating points

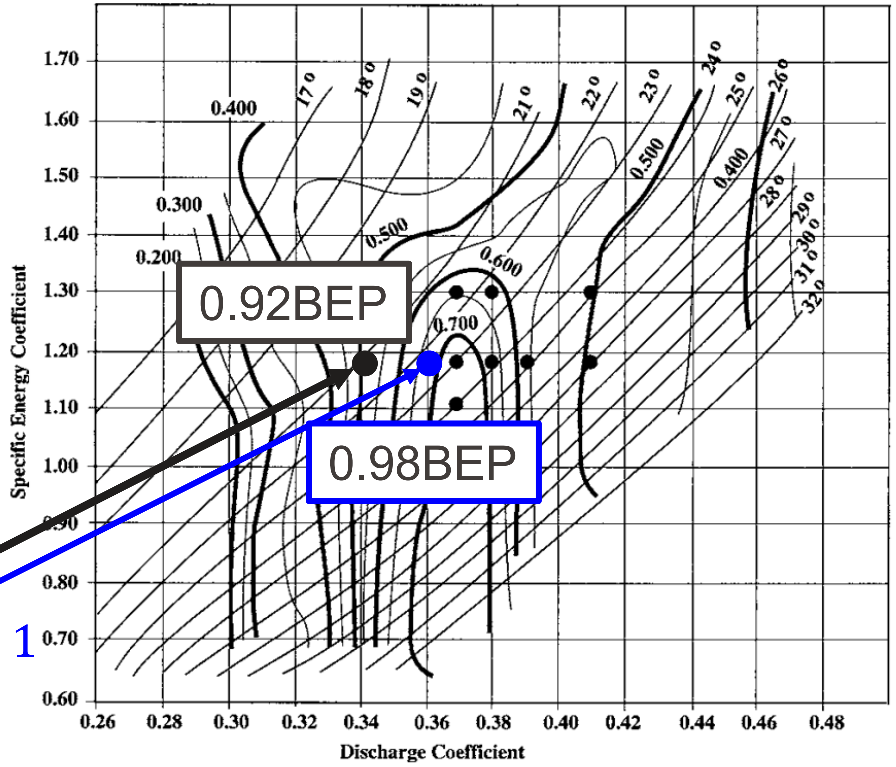
$$\boldsymbol{u}_{inlet} = (1-\alpha)\,\boldsymbol{u}_{inlet,\,0.92BEP} + \alpha\,\boldsymbol{u}_{inlet,\,0.98BEP}$$



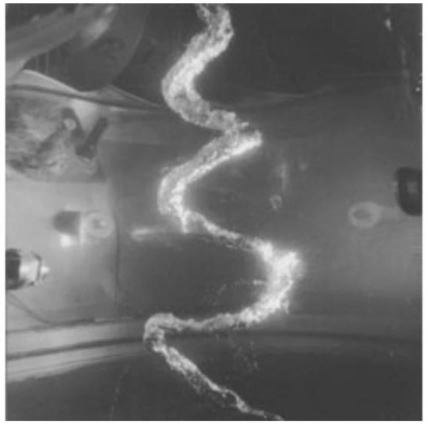


$\alpha = 0$

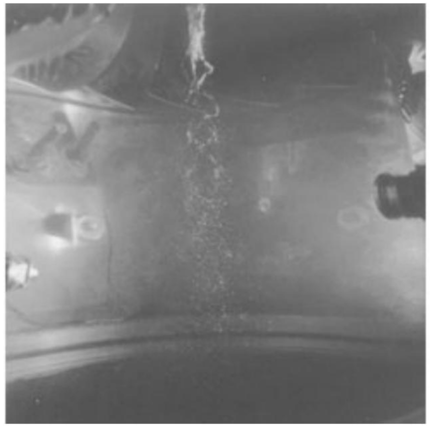
$\alpha = 1$



0.92 BEP



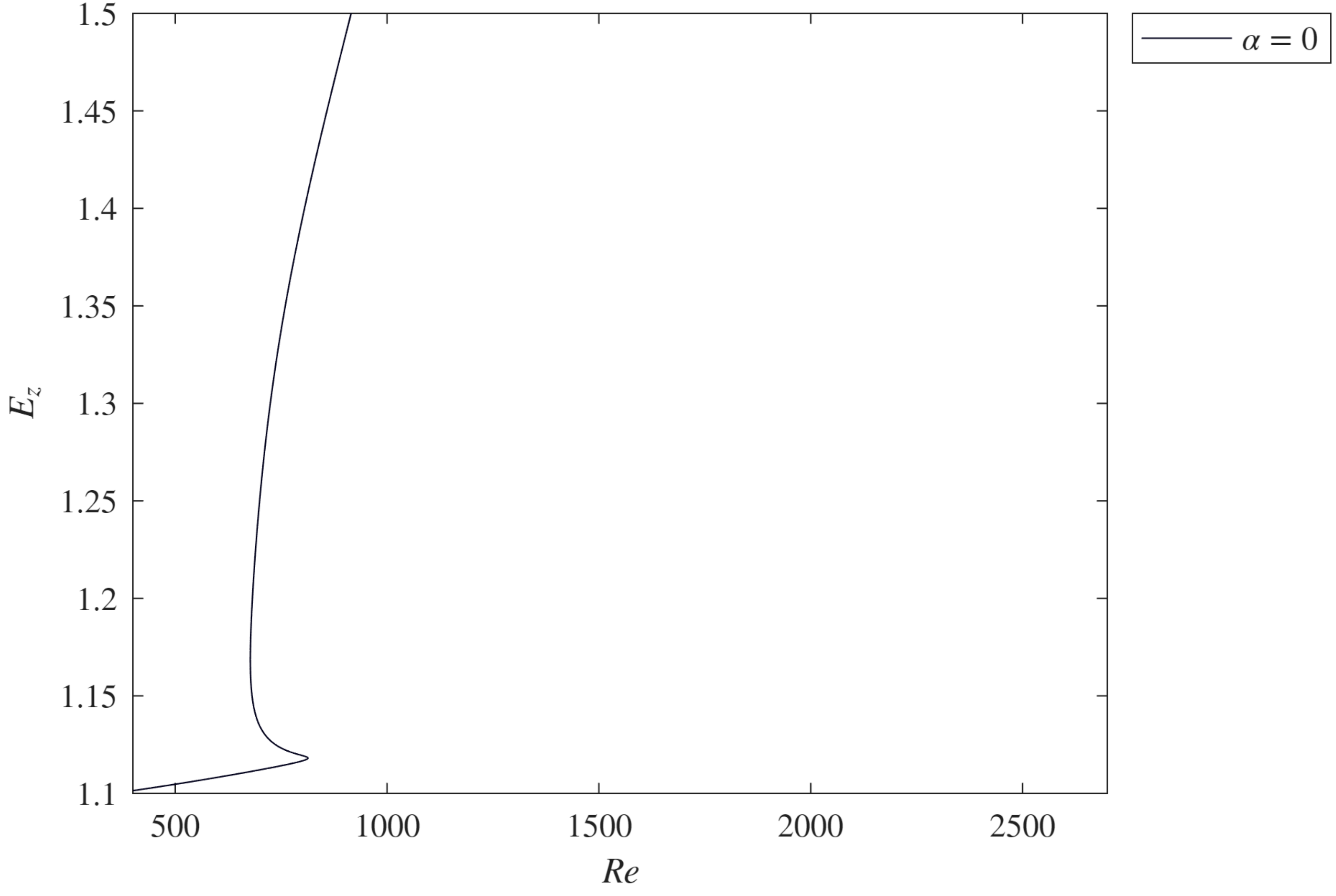
0.98 BEP



Mauri, PhD 2002

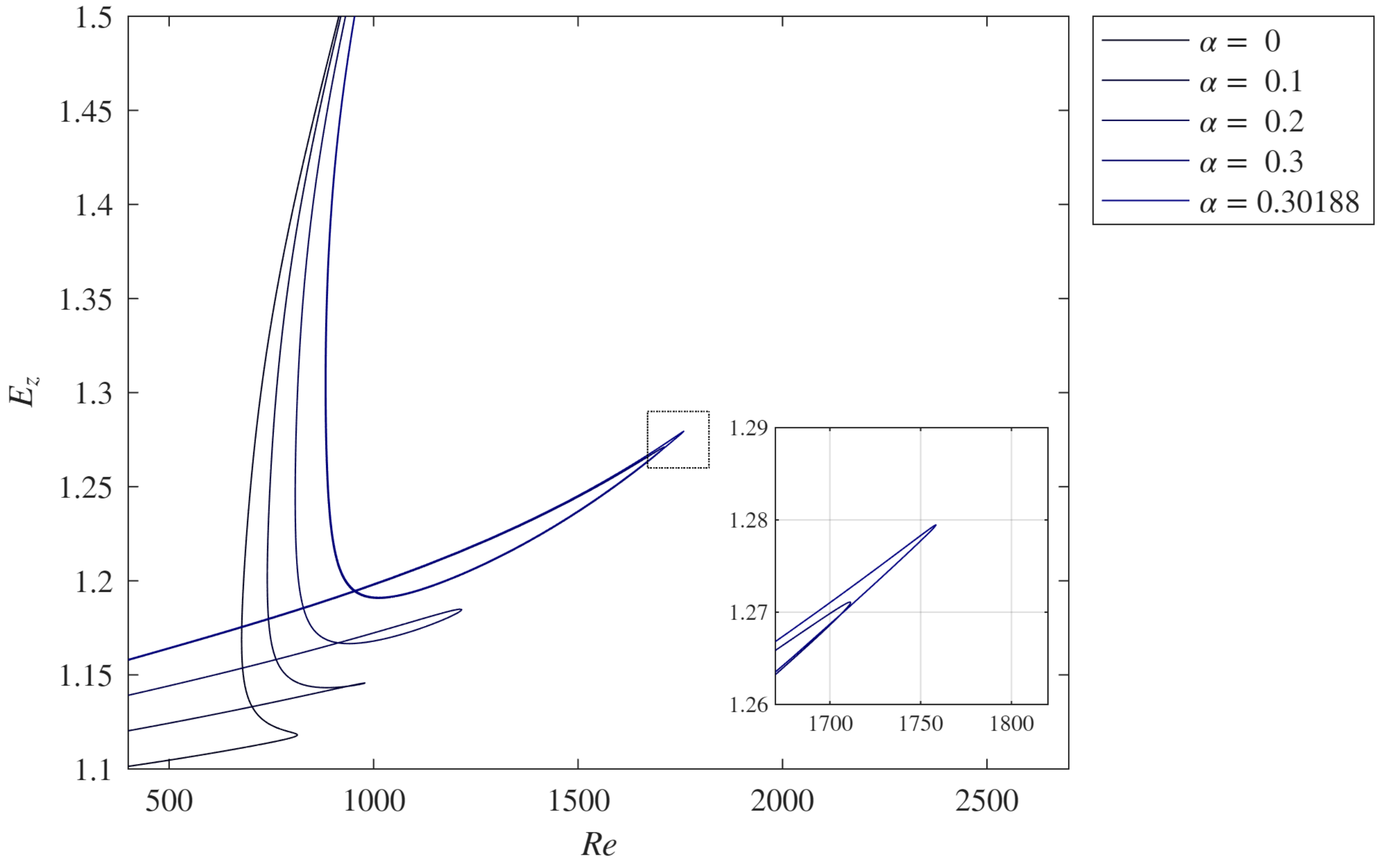
Slip wall

$$\mathbf{u}_{inlet} = \mathbf{u}_{inlet,\varphi=0.34}(1-\alpha) + \alpha\ \mathbf{u}_{inlet,\varphi=0.36}$$



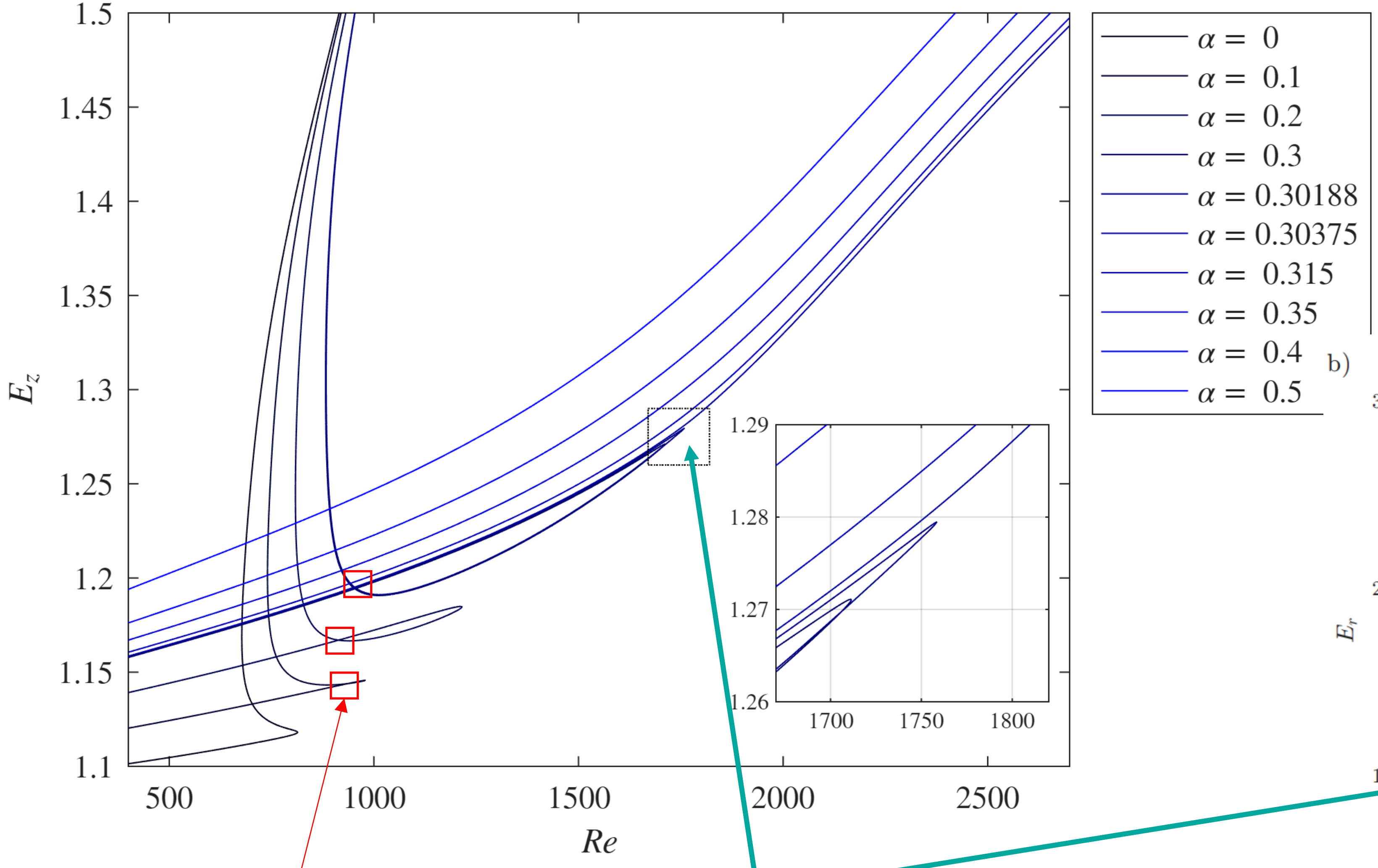
$$E_z = \frac{1}{2}\int_0^{15}\int_0^{2\pi}\int_0^{R(z)} u_z^2\ r\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}z$$

$$\mathbf{u}_{inlet} = \mathbf{u}_{inlet,\varphi=0.34}(1-\alpha) + \alpha\,\mathbf{u}_{inlet,\varphi=0.36}$$

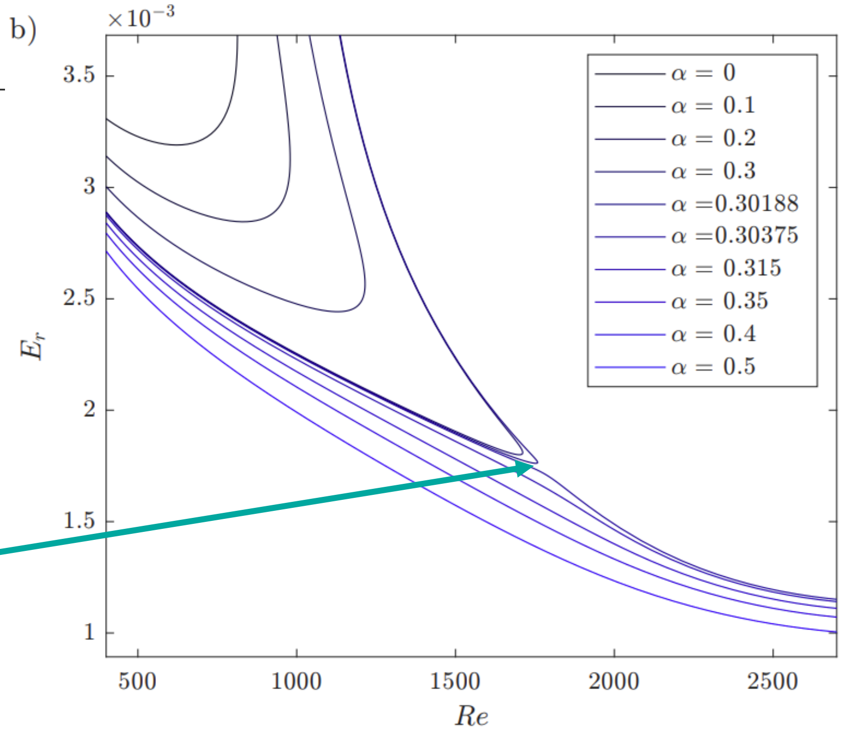


$$E_z = \frac{1}{2}\int_0^{15}\int_0^{2\pi}\int_0^{R(z)} u_z^2 \, r \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}z$$

$$\mathbf{u}_{inlet} = \mathbf{u}_{inlet,\varphi=0.34}(1-\alpha) + \alpha \, \mathbf{u}_{inlet,\varphi=0.36}$$



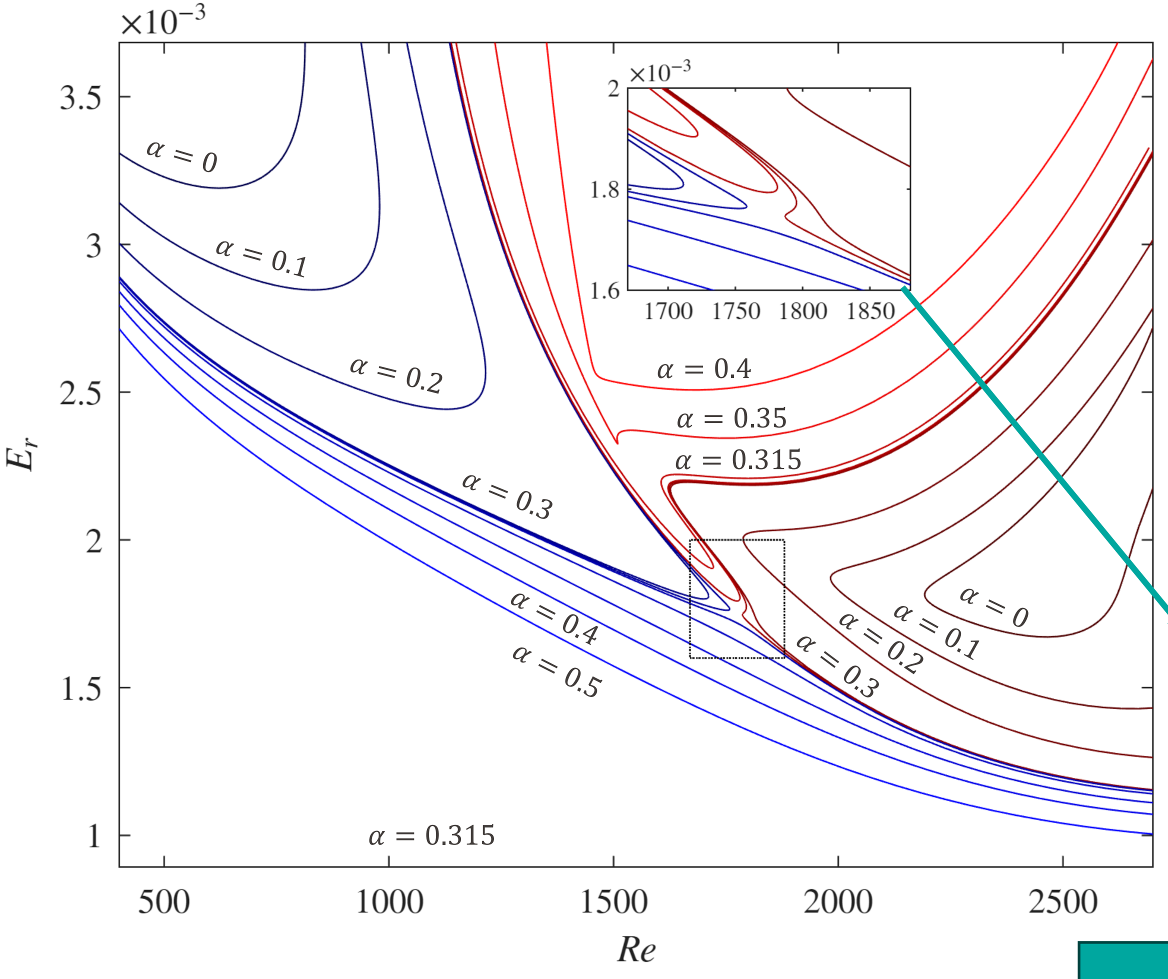
$$E_r = \frac{1}{2}\int_0^{15}\int_0^{2\pi}\int_0^{R(z)} u_r^2 \, r \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}z$$

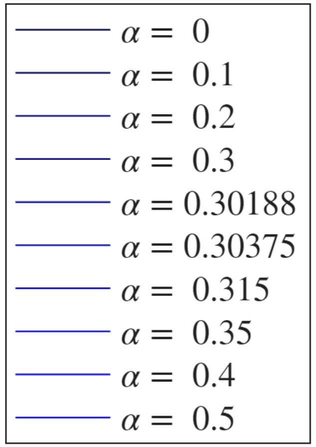


Self-intersection

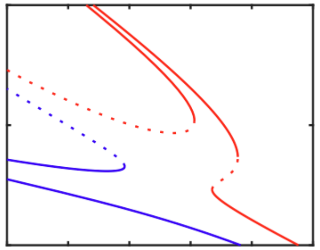
$\alpha_c = 0.3028$ critical point

Additional branches



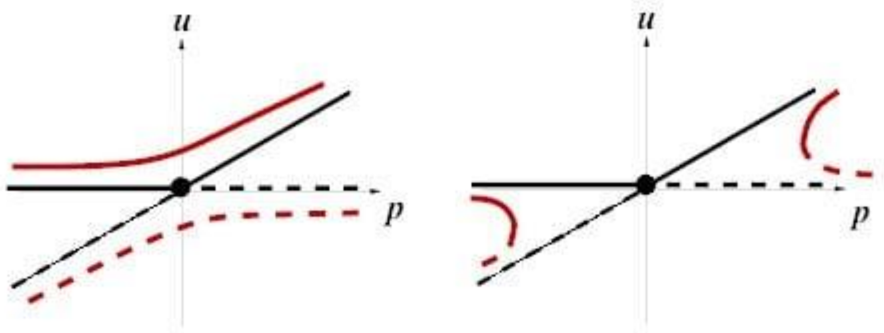


Stability m=0

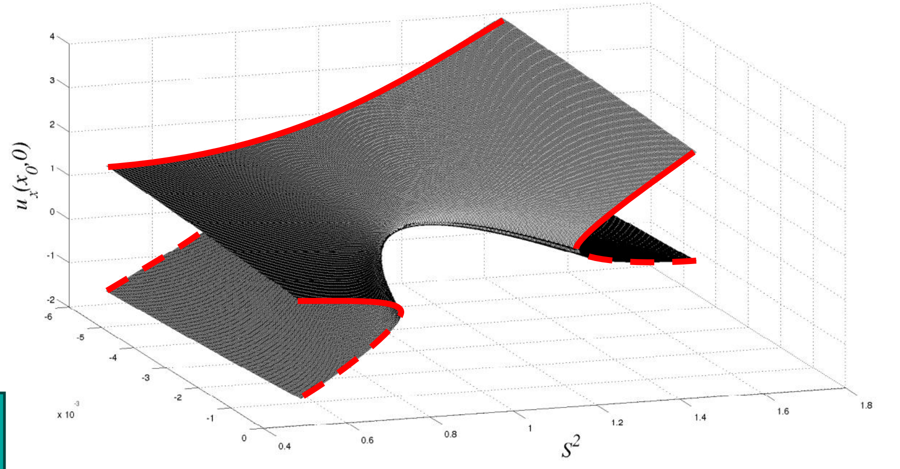


Transcritical bifurcation

Perturbed transcritical bifurcation



Simonnet *et al.*, Handb. Numer. Anal. 2009



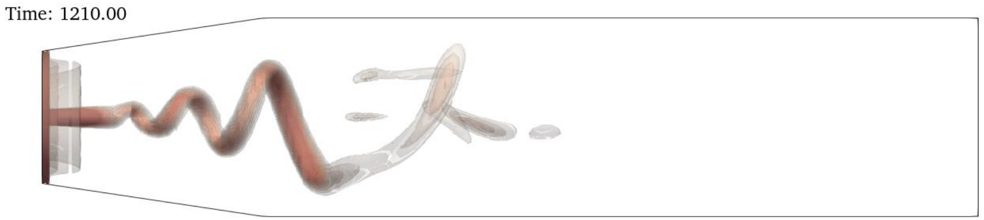
Vyazmina, Bifurcations in a swirling flow, PhD 2010

Summary

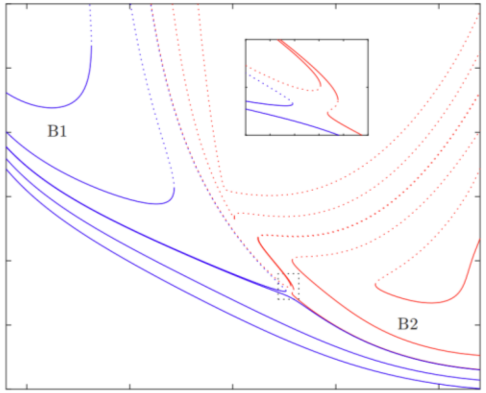
Laminar vortex rope computations :

- No-slip wall – Hopf bifurcation

- Free slip wall – breathing dynamics

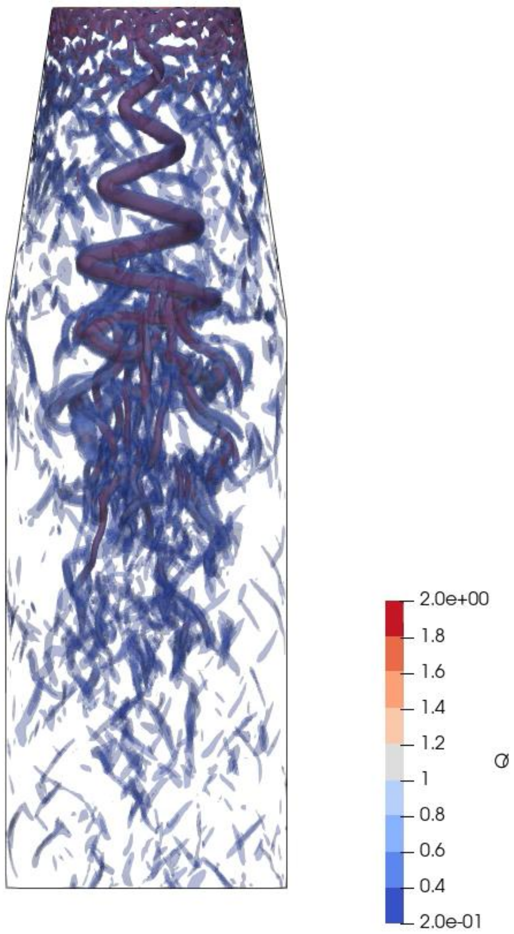


- Transcritical bifurcation (free slip wall)



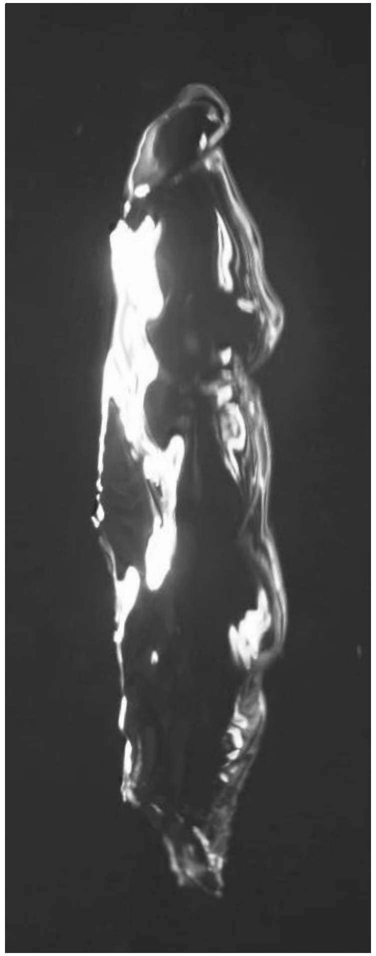
Perspectives

Turbulent breakdown



LES (openFoam)

Two-phase breakdown



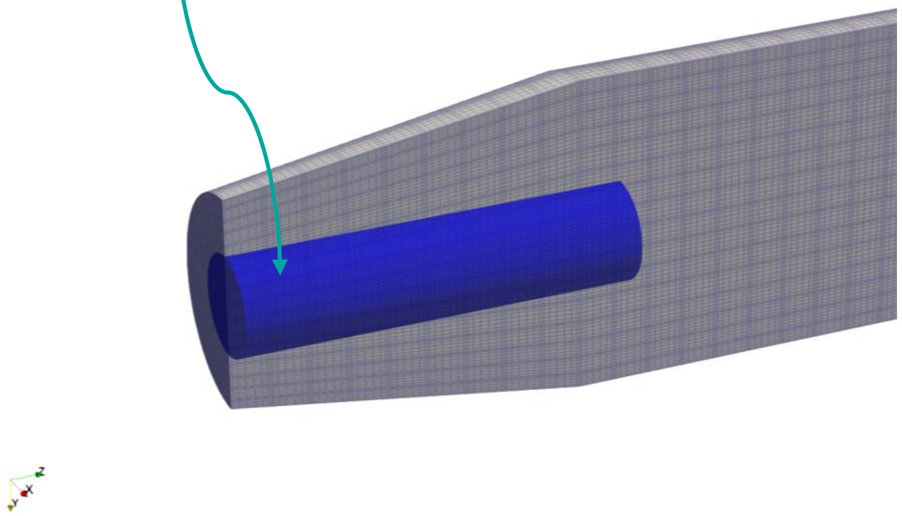
P. Solis *et al.*, JFM 2026

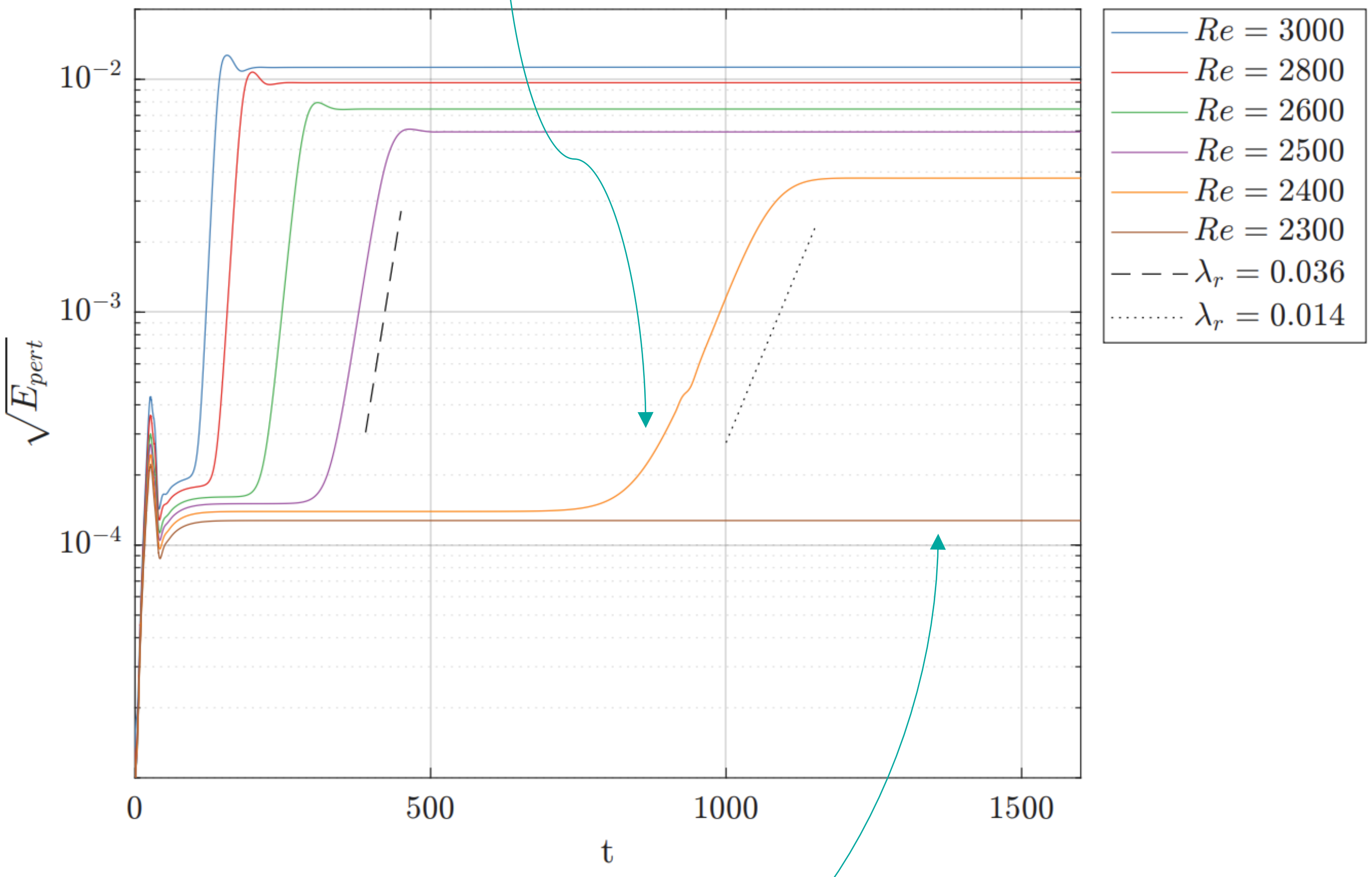
Extras

Departure from base solution at Re>2300.

Time integration

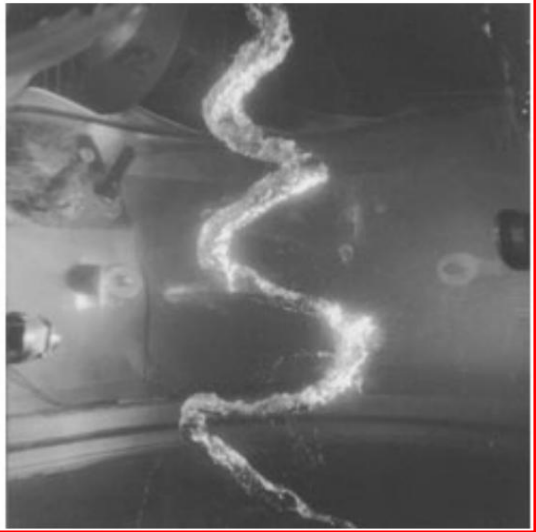
$$E_{pert} = \frac{1}{2}\int_0^4 \int_0^{2\pi} \int_0^{0.5} u_r^2 + u_\theta^2 + u_z^2 \; r \, dr \, d\theta \, dz$$



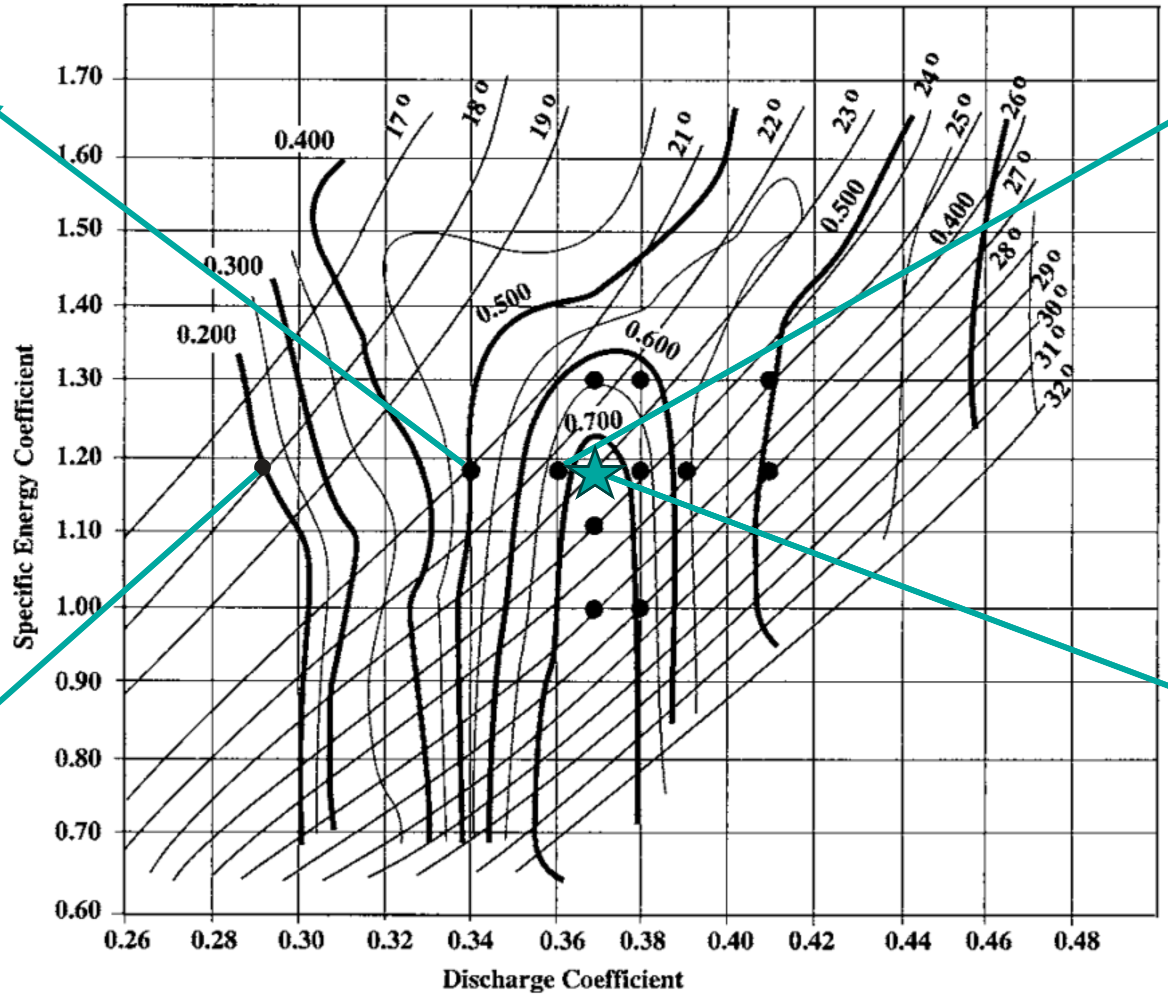


Flow stable at Re=2300 (as predicted by the linear stability analysis).

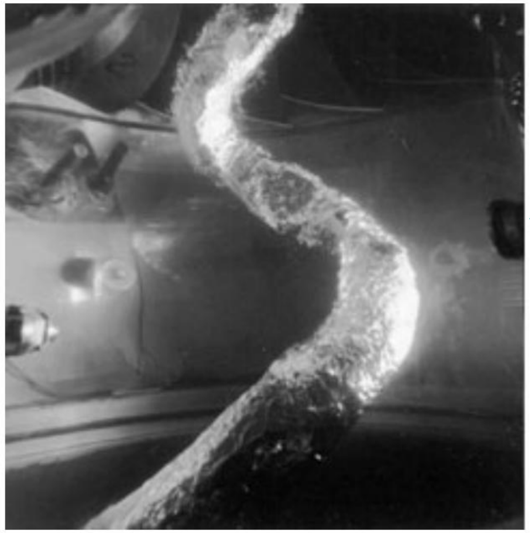
0.90 BEP Best efficiency point (1.0 BEP★)



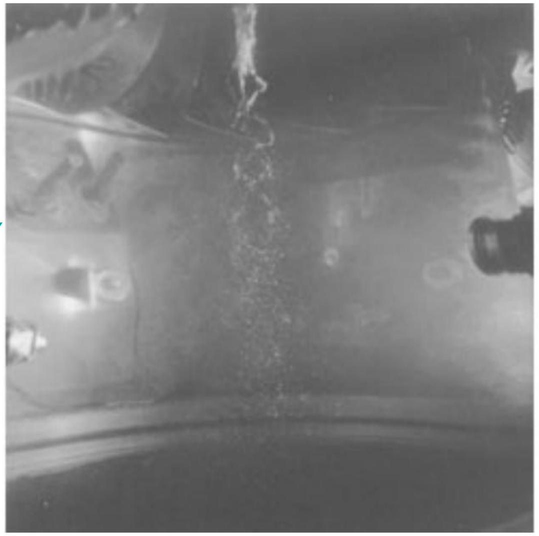
Partial load vortex rope



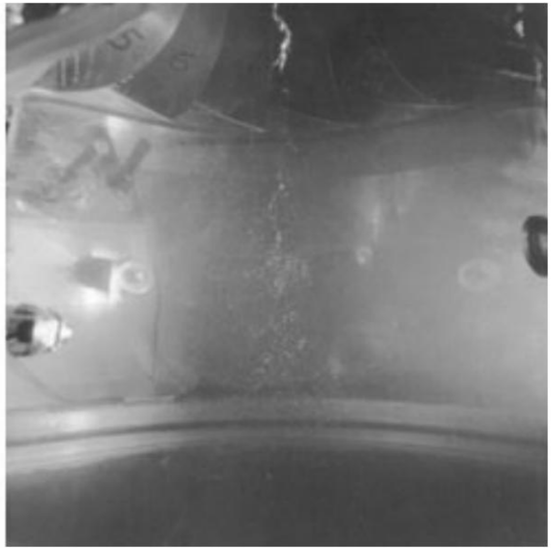
0.80 BEP



0.95 BEP

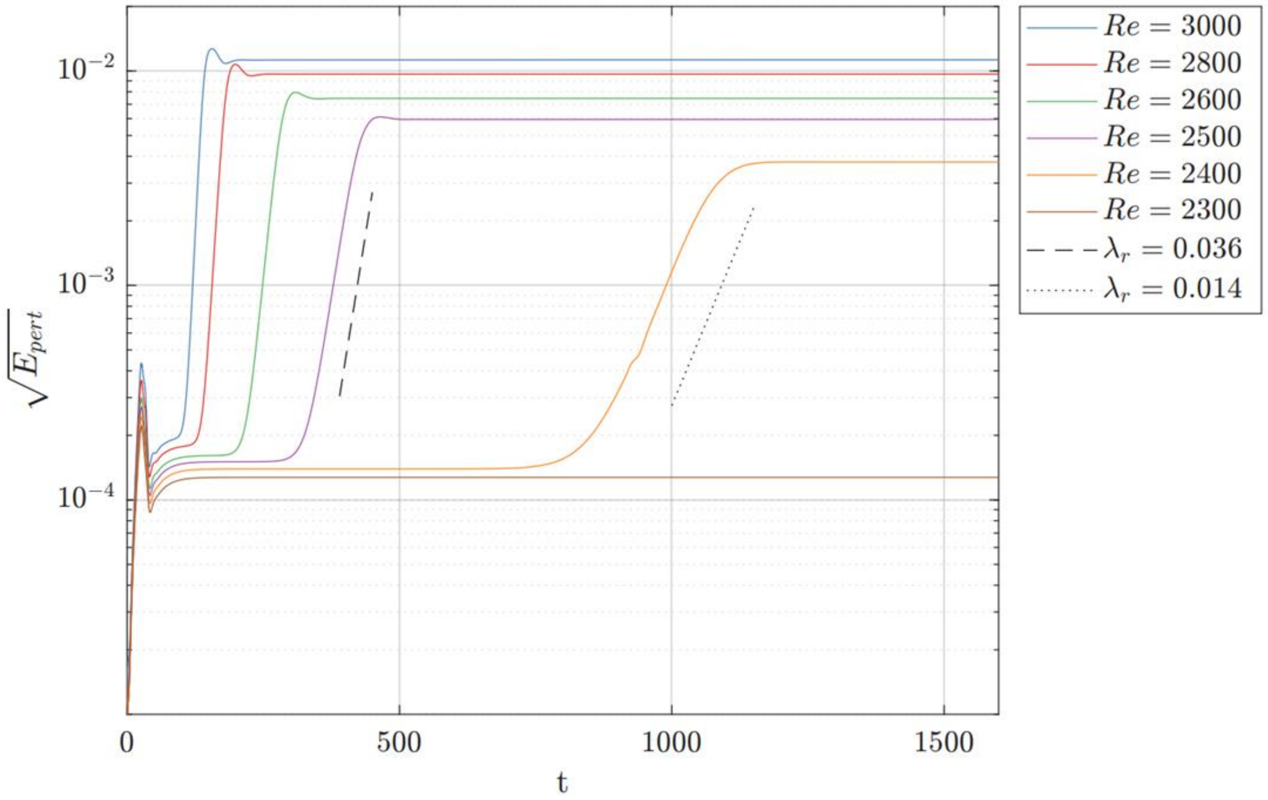


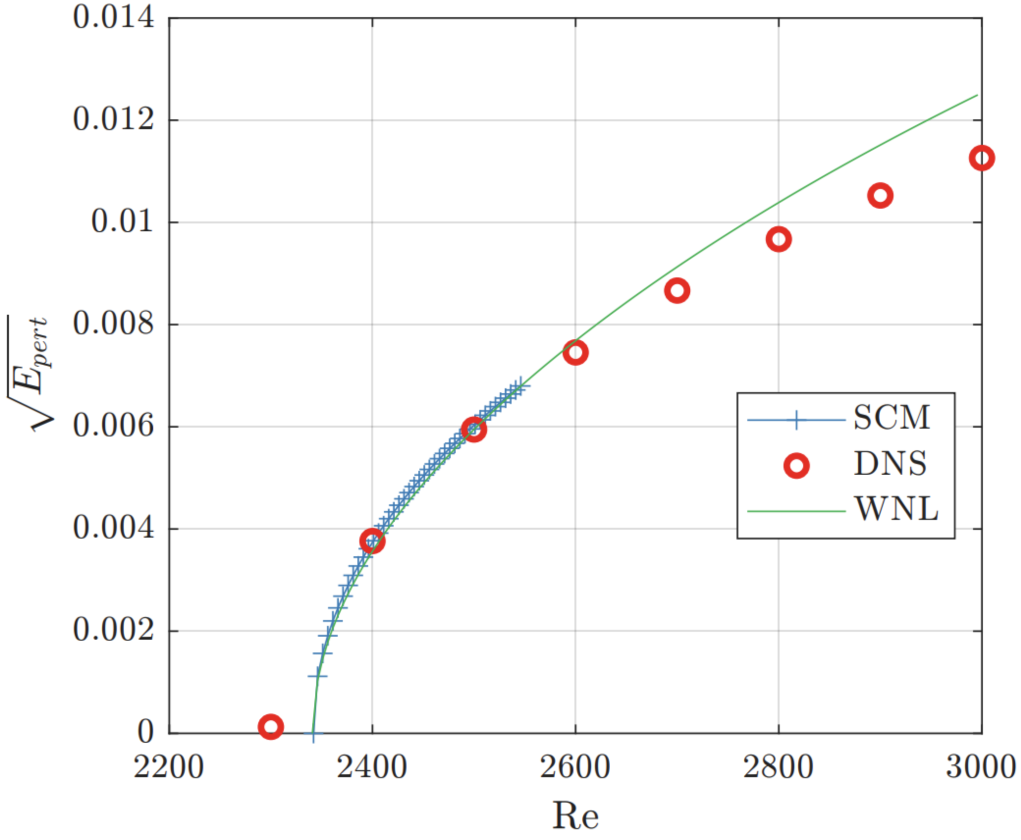
1.0 BEP



Jacob, PhD 1994; Mauri, PhD 2002; Pasche, PhD 2018;

Saturation level





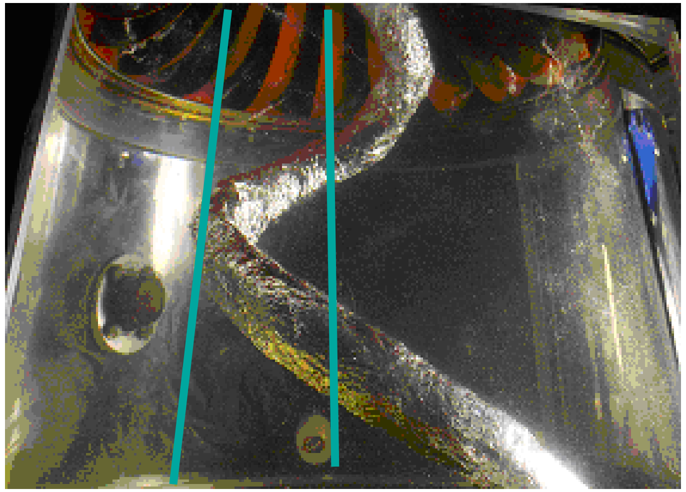
WNL – Weakly nonlinear analysis (e.g. Sipp & Lebedev, JFM 2007)
SCM – Self-Consistent Model (Mantič-Lugo *et al.*, PRL 2014)

EPFL

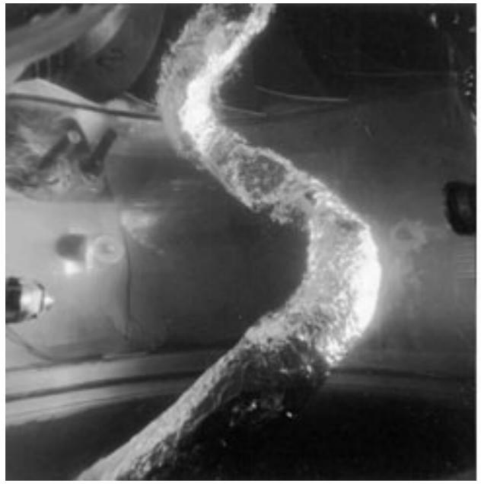
No-slip computations summary

Laminar no-slip wall simulation (Re ≈ 10^3)

- Vortex rope due to a supercritical Hopf bifurcation
- Rope frequency ≈ Ω
- No rope cone opening angle
- Wall receptivity mode
- No clear stagnation region at the axis



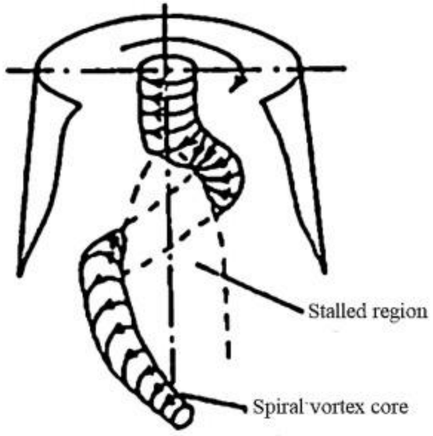
Ciocian *et al.*, JFE 2007



Mauri, PhD 2002

Experiment (Re=10^6)

- Frequency $0.2\Omega - 0.4\Omega$
- Rope cone opening angle
- Stall region at the axis



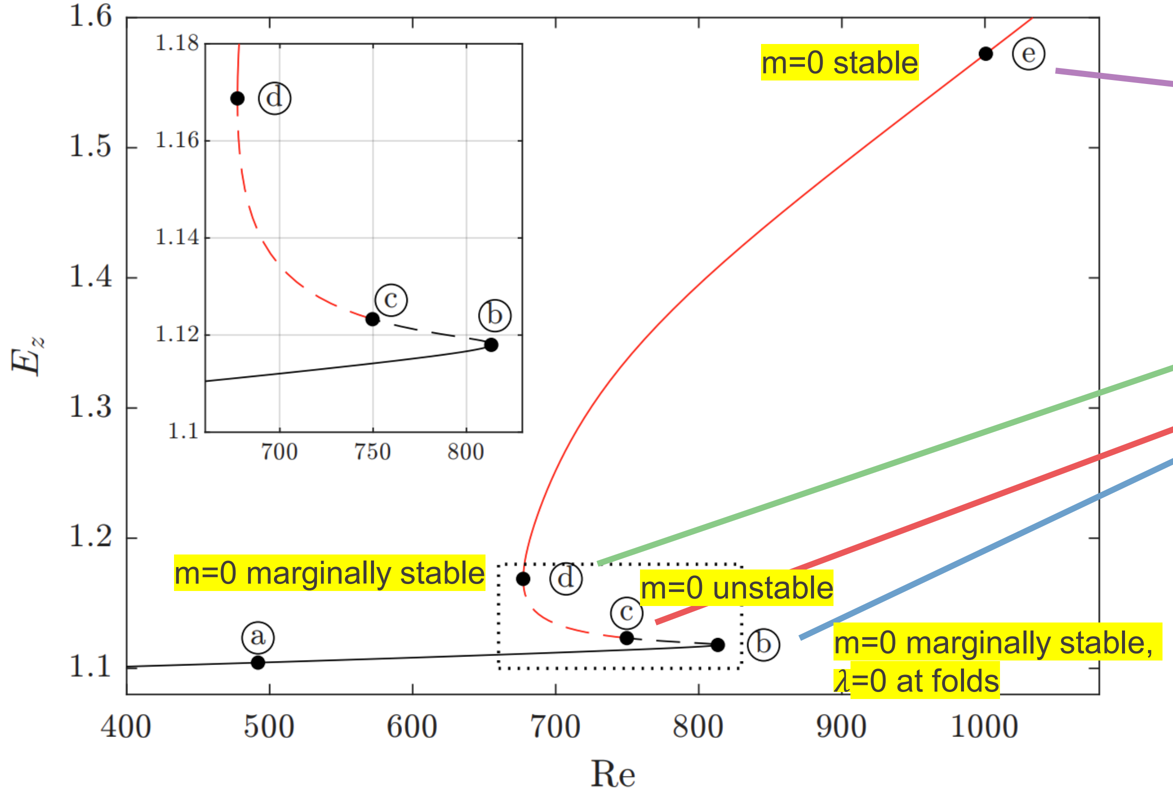
Nishi, IAHR Symp. 1982

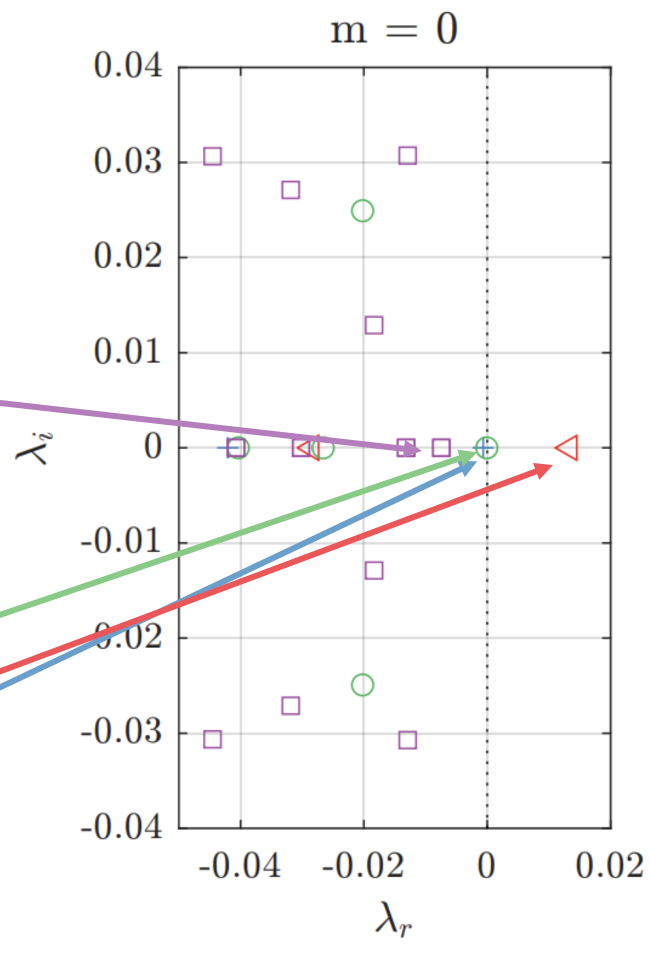
Maybe we could change the wall boundary condition to free slip.

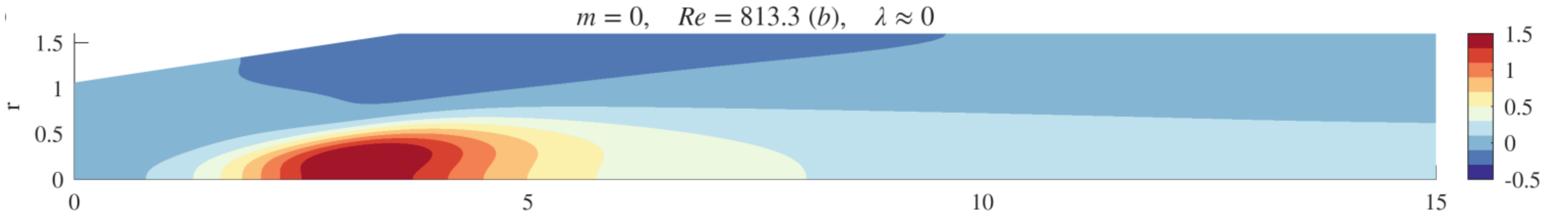
Lin. stab. analysis m = 0

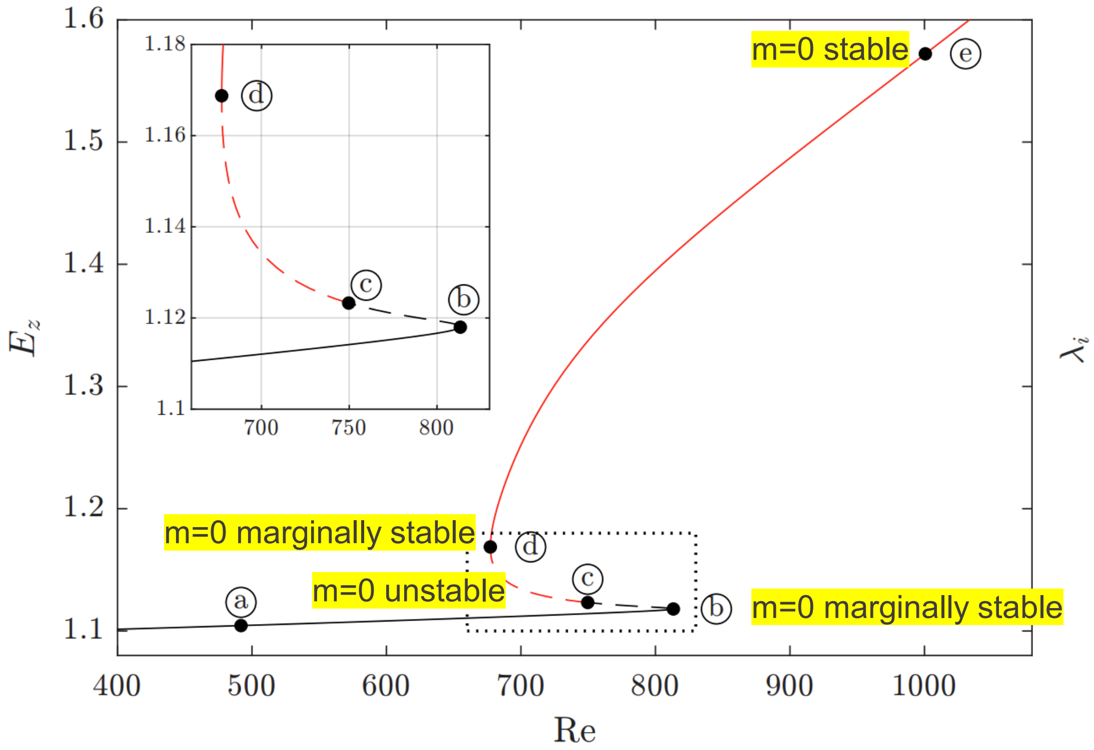
$u \sim \hat{u}(r,z)e^{im\theta+\lambda t}$

$\lambda_r > 0$ - unstable

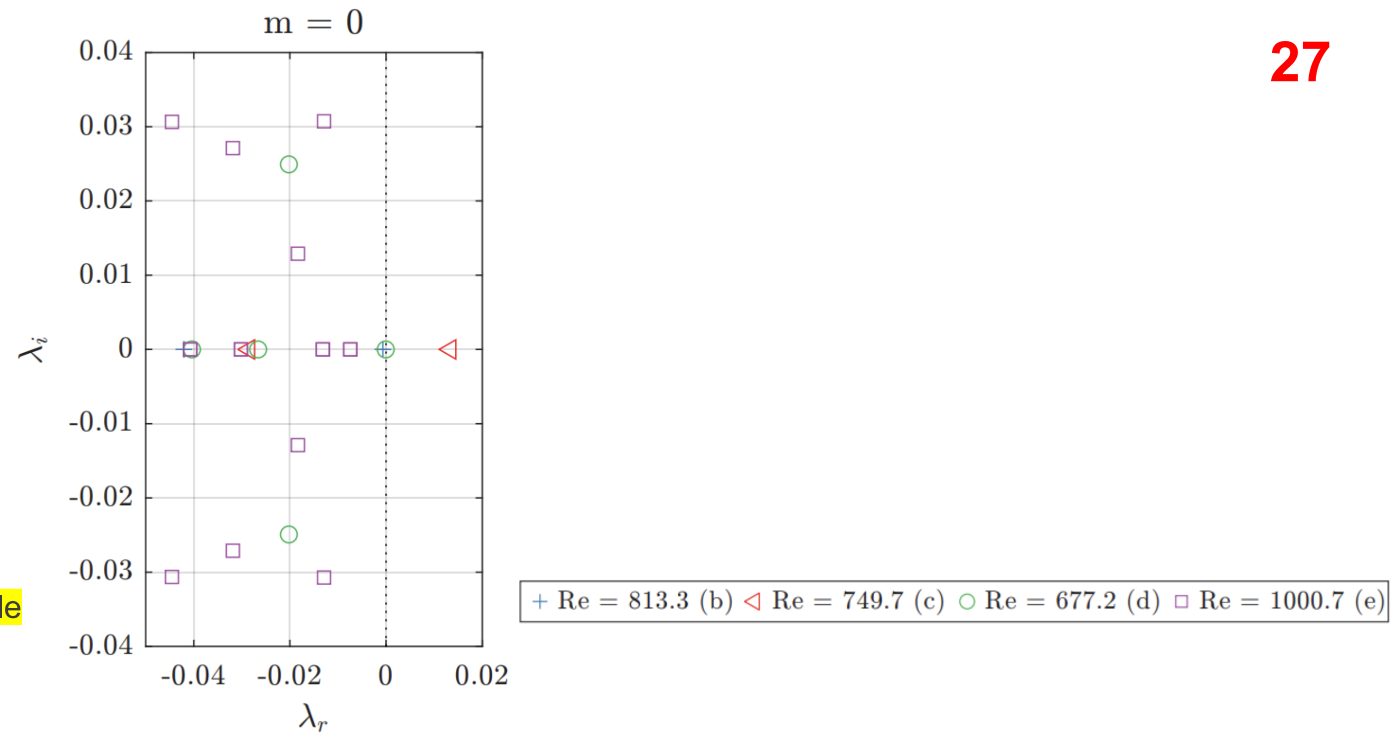




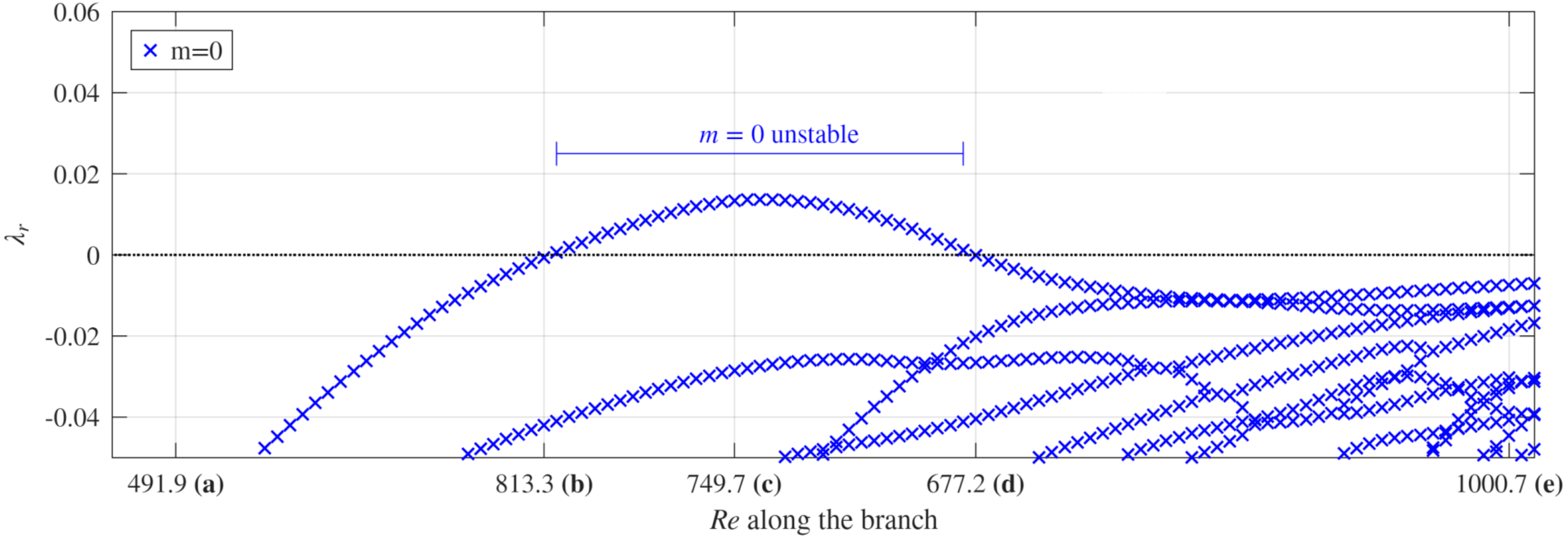




m=0 stable
m=0 marginally stable
m=0 unstable
m=0 marginally stable
(a)
(b)
(c)
(d)
(e)
E_z
Re
1.6
1.5
1.4
1.3
1.2
1.1
400
500
600
700
800
900
1000
1.18
1.16
1.14
1.12
1.1
700
750
800

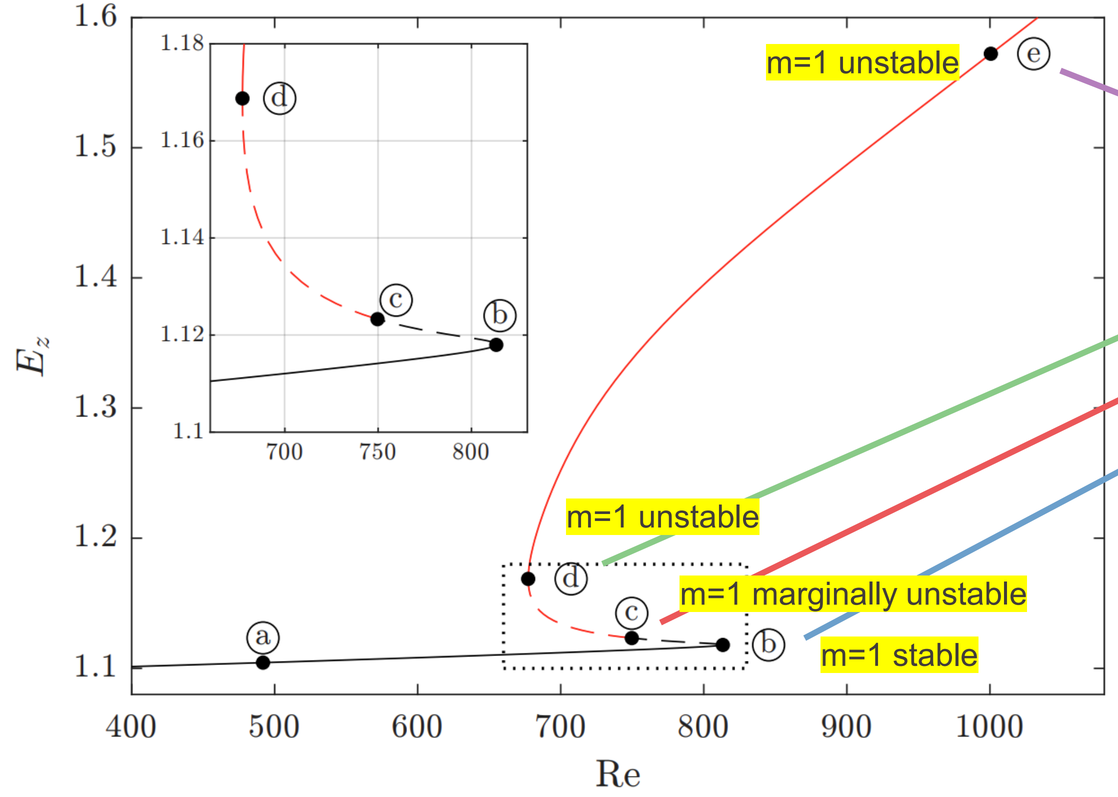


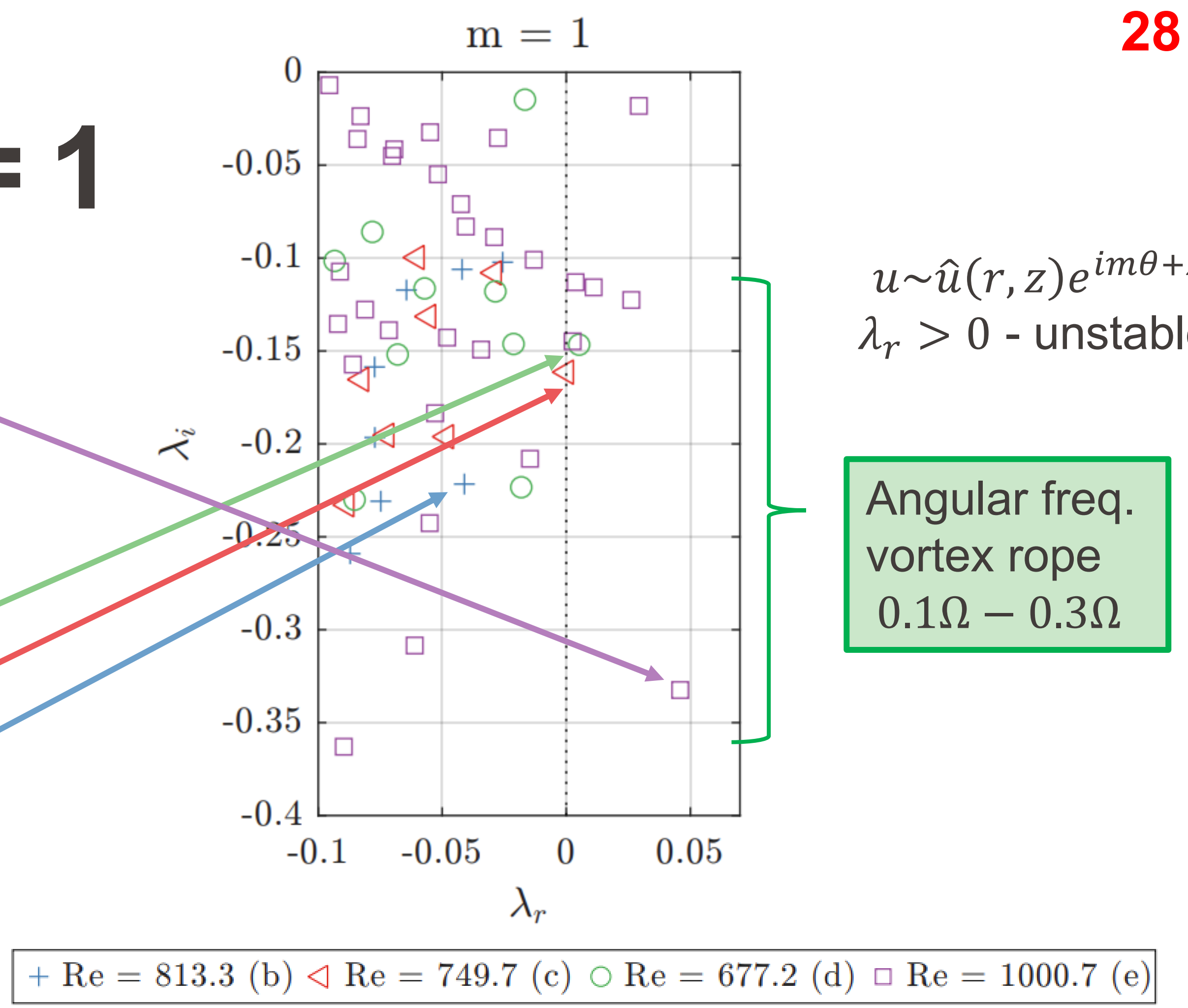
m = 0
λ_i
λ_r
0.04
0.03
0.02
0.01
0
-0.01
-0.02
-0.03
-0.04
-0.04
-0.02
0
0.02
+ Re = 813.3 (b)
◁ Re = 749.7 (c)
○ Re = 677.2 (d)
□ Re = 1000.7 (e)



× m=0
m = 0 unstable
λ_r
0.06
0.04
0.02
0
-0.02
-0.04
491.9 **(a)**
813.3 **(b)**
749.7 **(c)**
677.2 **(d)**
1000.7 **(e)**
Re along the branch

Lin. stab. analysis m = 1

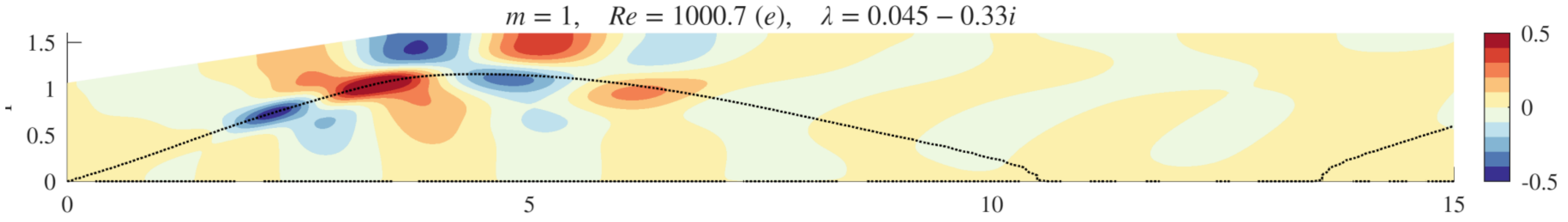


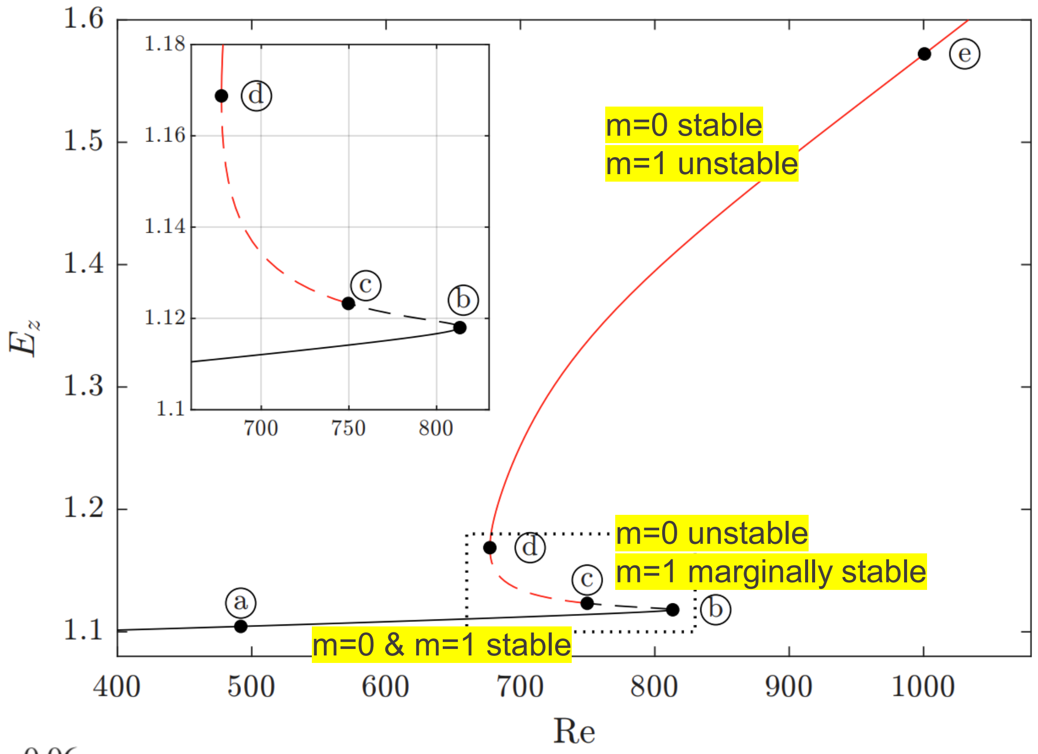


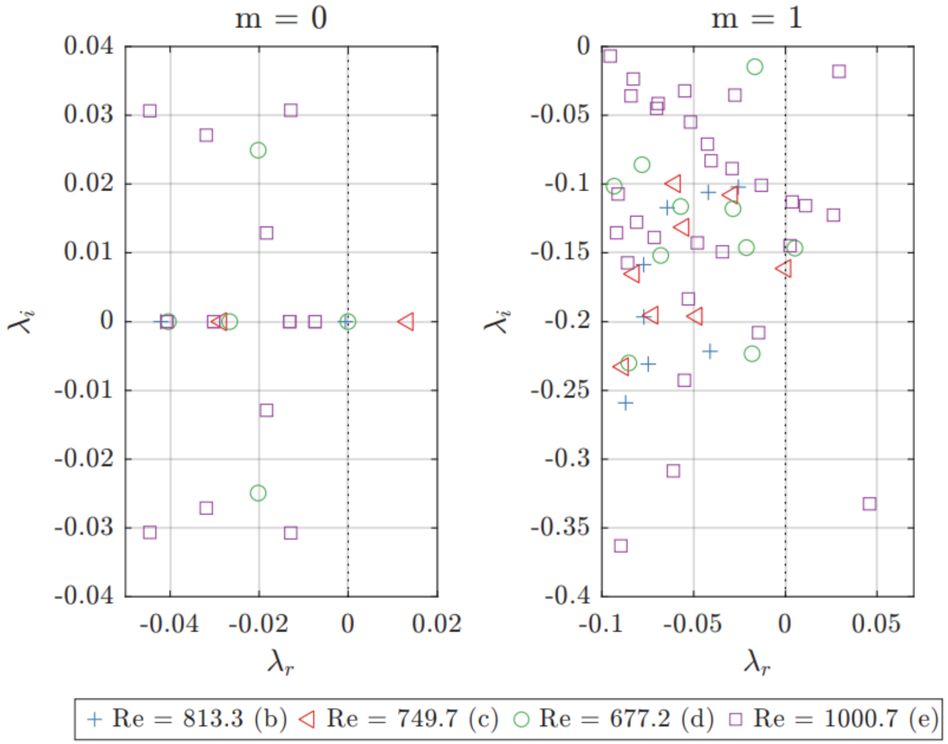
$u \sim \hat{u}(r,z)e^{im\theta+\lambda t}$

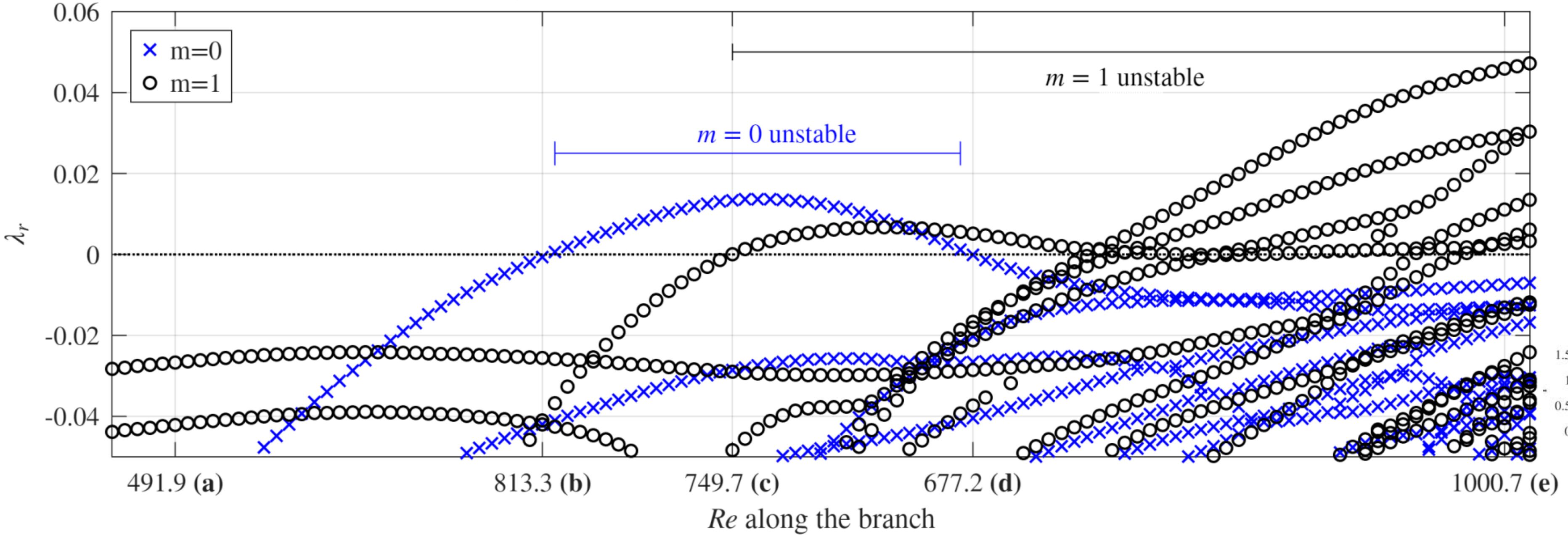
$\lambda_r > 0$ - unstable

Angular freq. vortex rope
$0.1\Omega - 0.3\Omega$



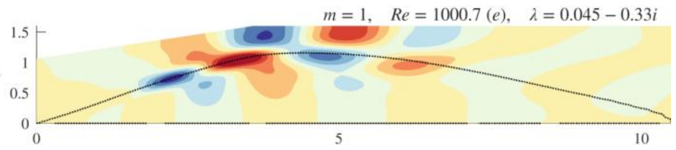




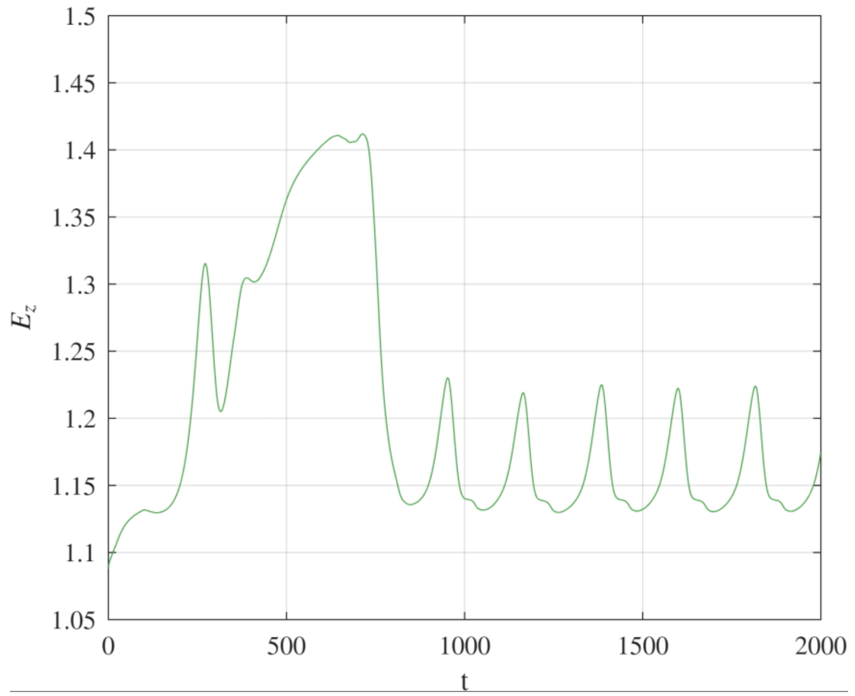


m=1 unstable mode :

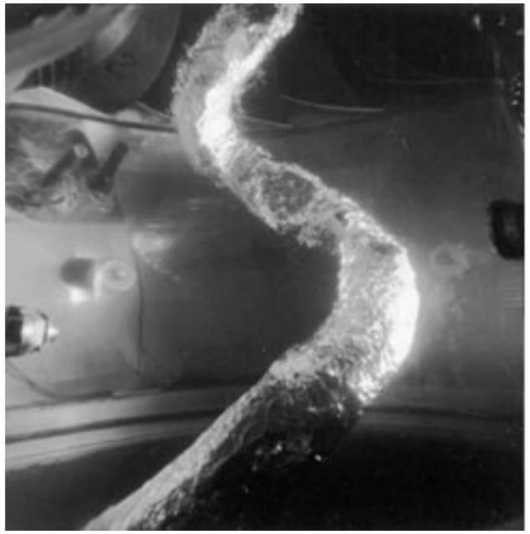
- spiral mode
- just the right frequency
- at the boundary of the recirculation bubble

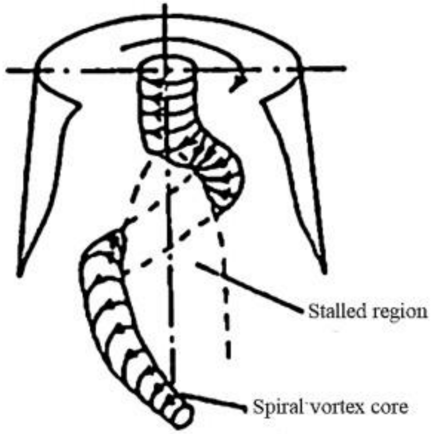


Re=840 - breathing



Summary - slip wall



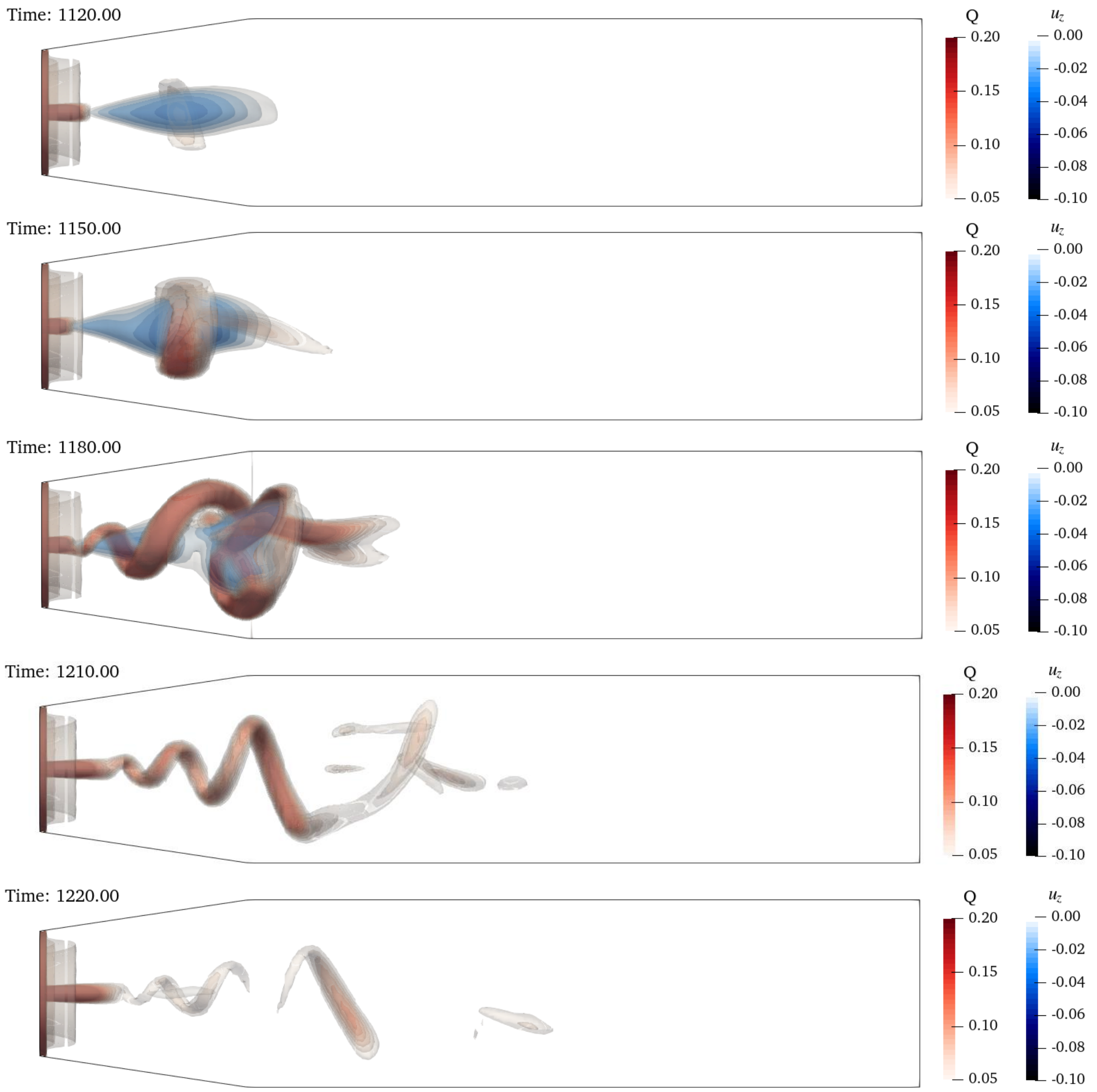


Laminar slip wall simulation

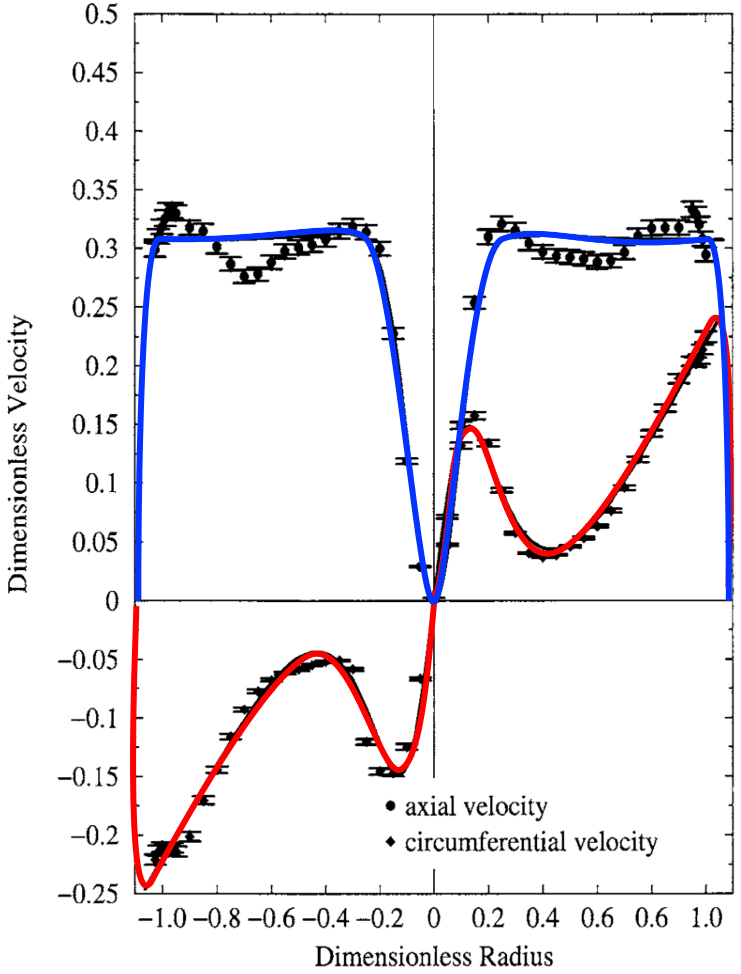
- Clear rope cone opening angle
- No boundary layer receptivity mode
- Clear recirculation bubble at the axis
- Rope frequency $\approx 0.2 - 0.4\Omega$

Like in the turbulent experiment.

Breathing dynamics



Inlet profile



Susan-Resiga *et al.*, JFE 2006

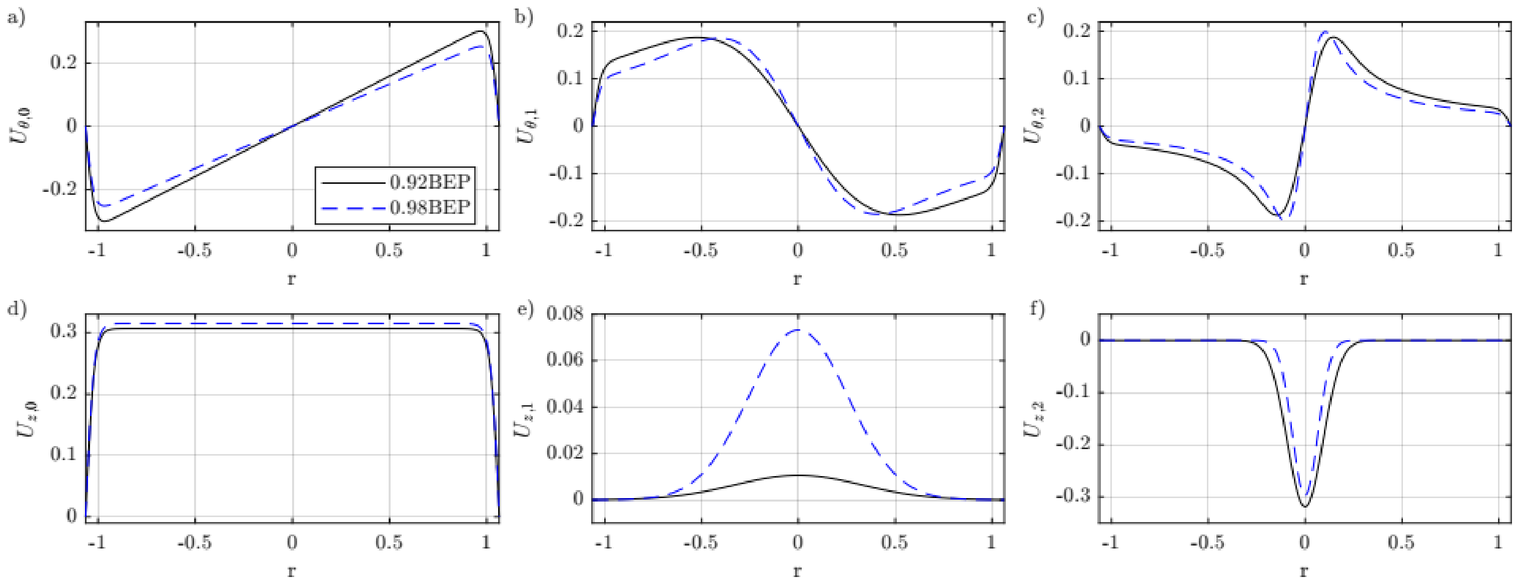


FIG. 3. Decomposition of the inlet profiles defined in equations (1)-(3) on three elementary components : a solid body rotation with a uniform axial flux and two counter rotating Batchelor vortices for two turbine operating points (cf. table I for the values of numerical parameters)

$$U_r = 0,$$

$$U_\theta = \Omega_0 r + \frac{\Omega_1 r_1^2}{r}\left[1 - \exp(-r^2/r_1^2)\right] + \frac{\Omega_2 r_2^2}{r}\left[1 - \exp(-r^2/r_2^2)\right],$$

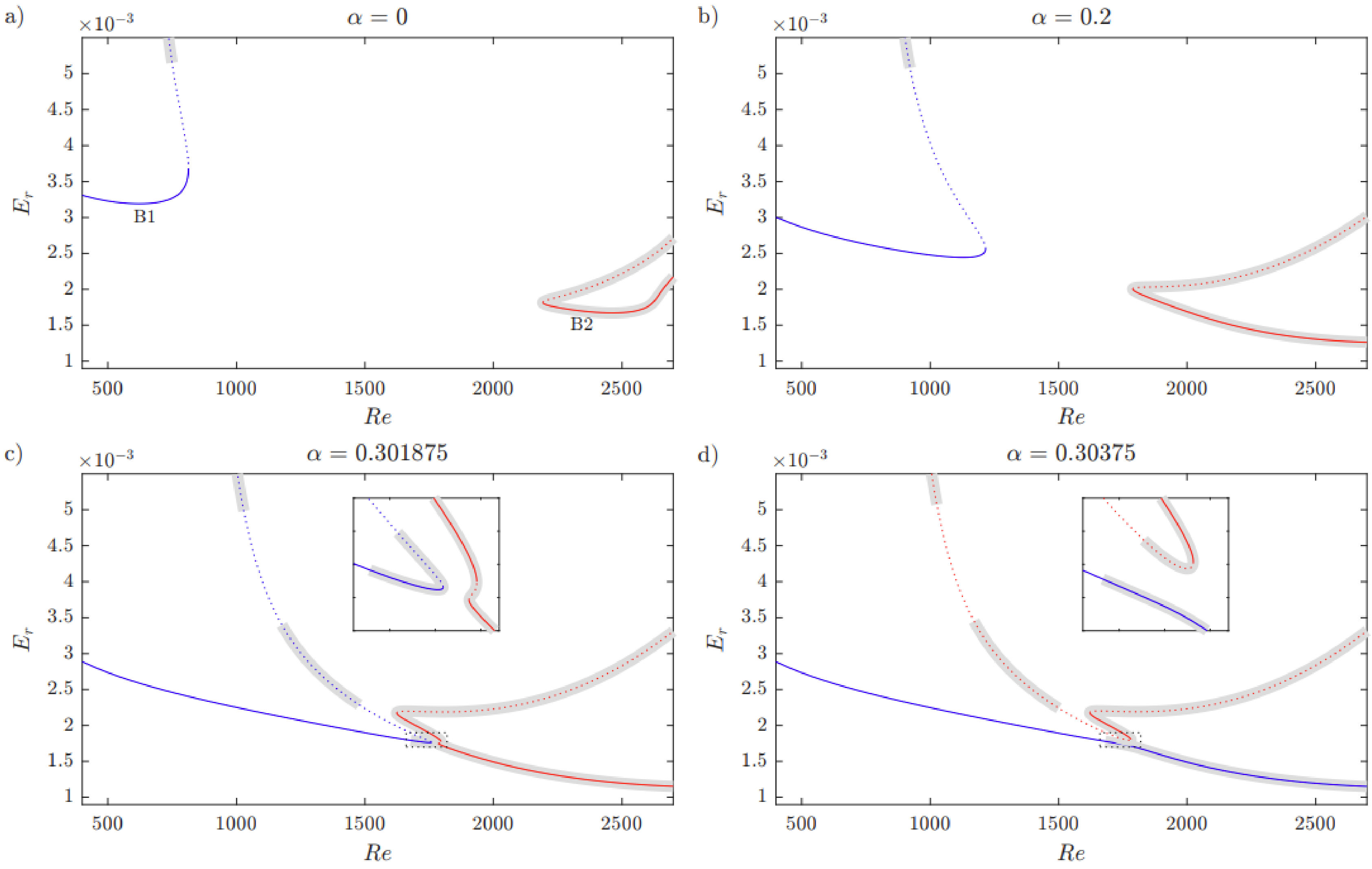
$$U_z = U_0 + U_1 \exp(-r^2/r_1^2) + U_2 \exp(-r^2/r_2^2).$$

4

φ	BEP	Ω_0	Ω_1	Ω_2	U_0	U_1	U_2	r_1	r_2
0.34	0.92	0.31765	-0.62888	2.2545	0.30697	0.01056	-0.31889	0.46643	0.13051
0.36	0.98	0.26675	-0.79994	3.3512	0.31501	0.07324	-0.29672	0.36339	0.09304

TABLE I. Inflow profile parameters extracted from [51]. The discharge coefficient φ is defined in [51] and is reproduced here for convenience.

Stability of slip branches



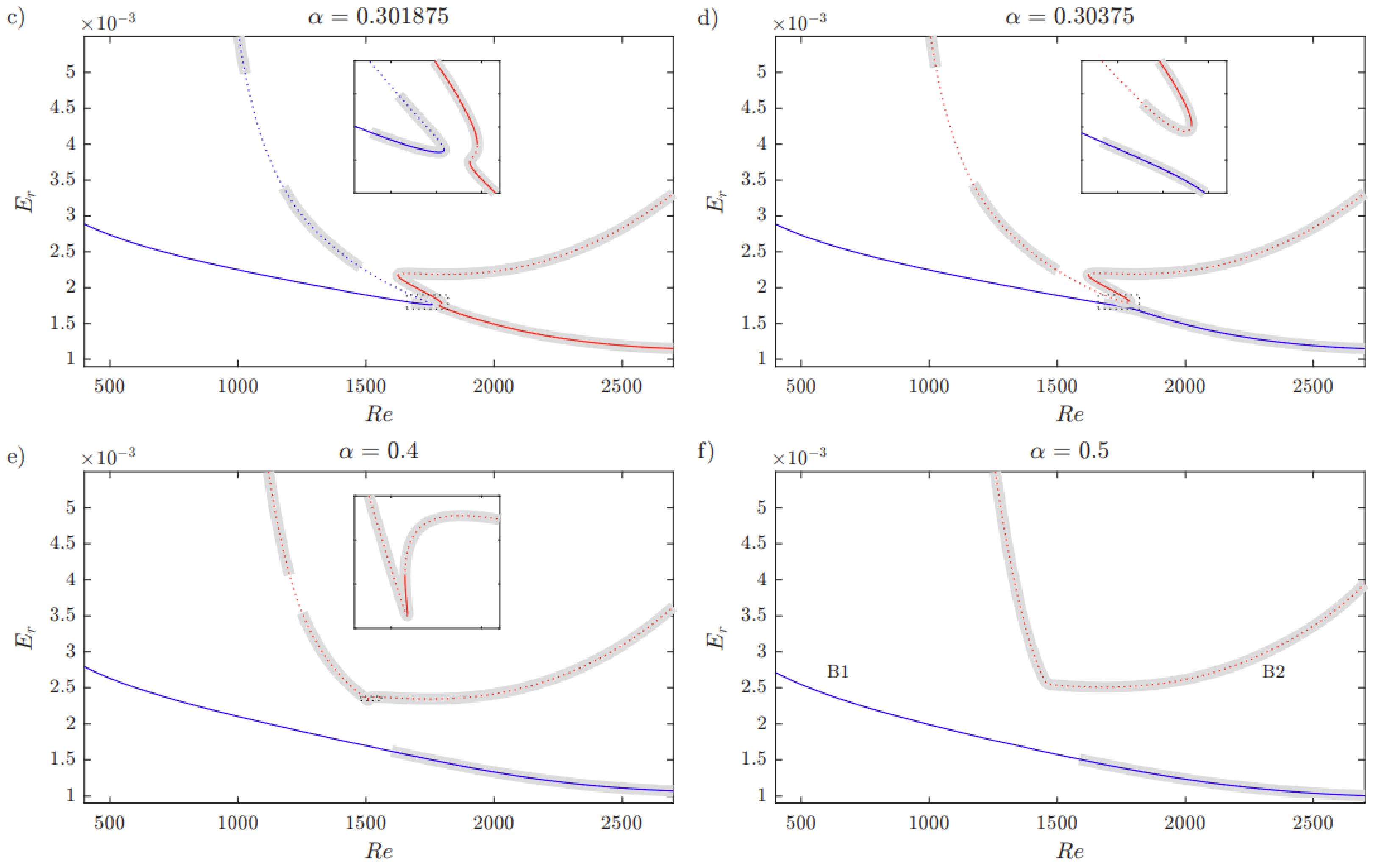
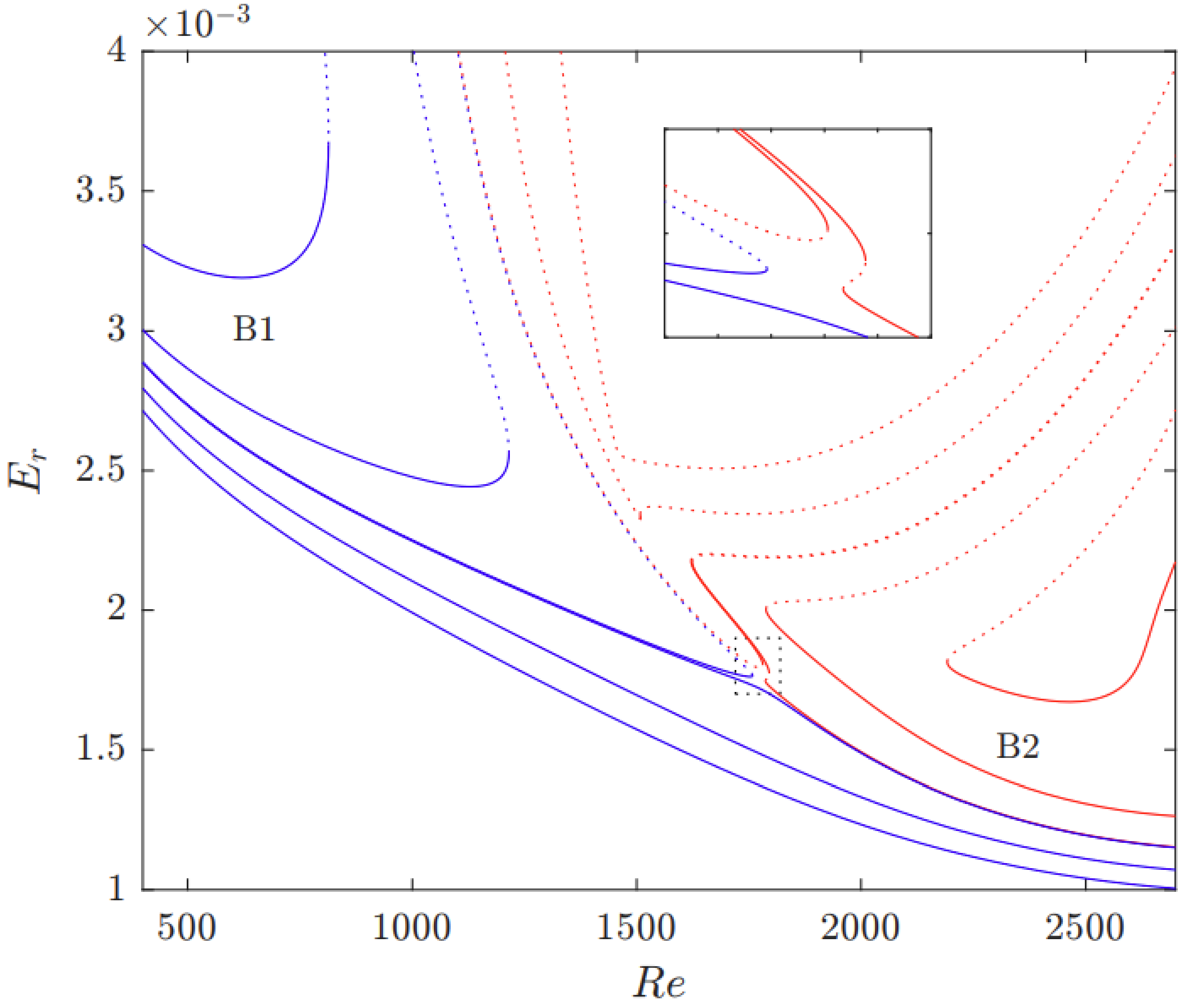


FIG. 18. Two branches of steady solutions identified with arclength continuation procedure for a range of interpolation parameters α. Collision of the branches between points $\alpha = 0.301875$ and $\alpha = 0.30375$ (panels c-d) marks a transcritical bifurcation. Insets in the corresponding panels correspond to a region bordered by a black dotted line magnified for better clarity. Stability to $m = 0$ modes is marked with solid (stable) and dotted (unstable) lines. Instability to $m = 1$ modes in marked with a shaded region.



$\times 10^{-3}$
4
3.5
3
2.5
2
1.5
1
E_r
500
1000
1500
2000
2500
Re
B1
B2

J. Fluid Mech. (1983), *vol.* 126, *pp.* 335–356
Printed in Great Britain

A sufficient condition for the instability of columnar vortices

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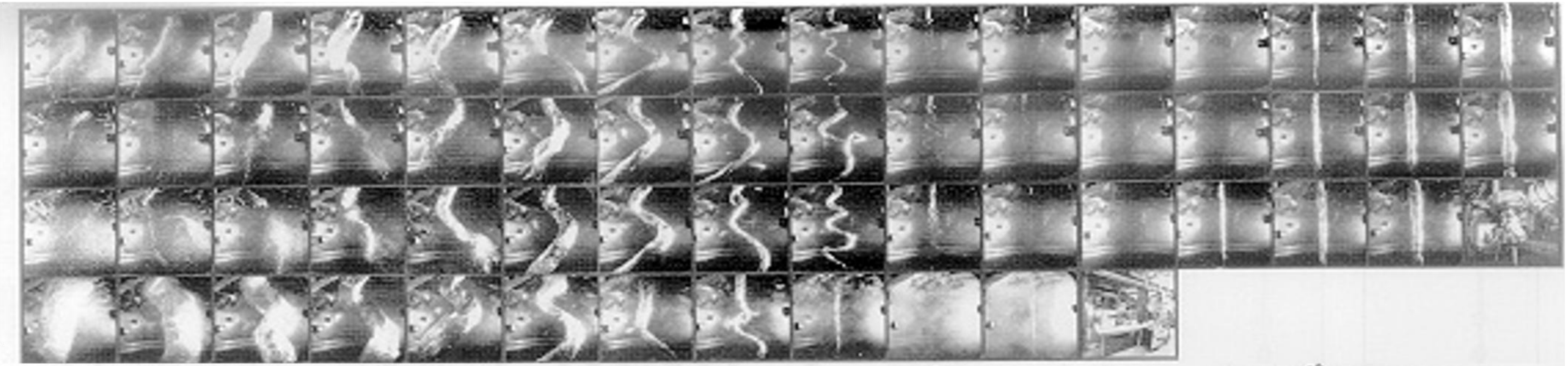
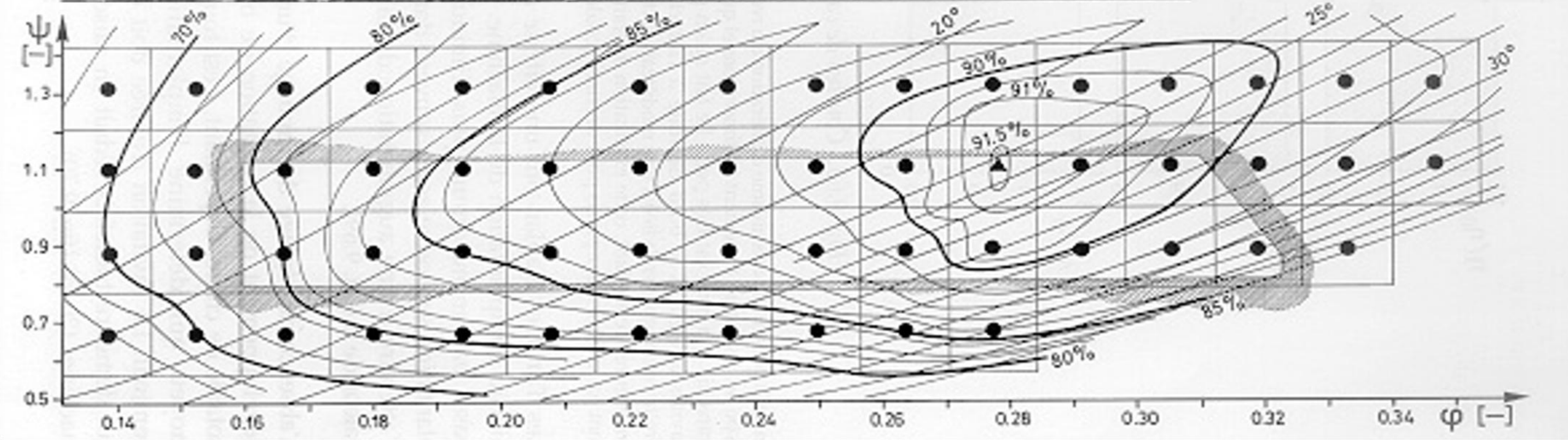
AND K. STEWARTSON
Department of Mathematics, University College London

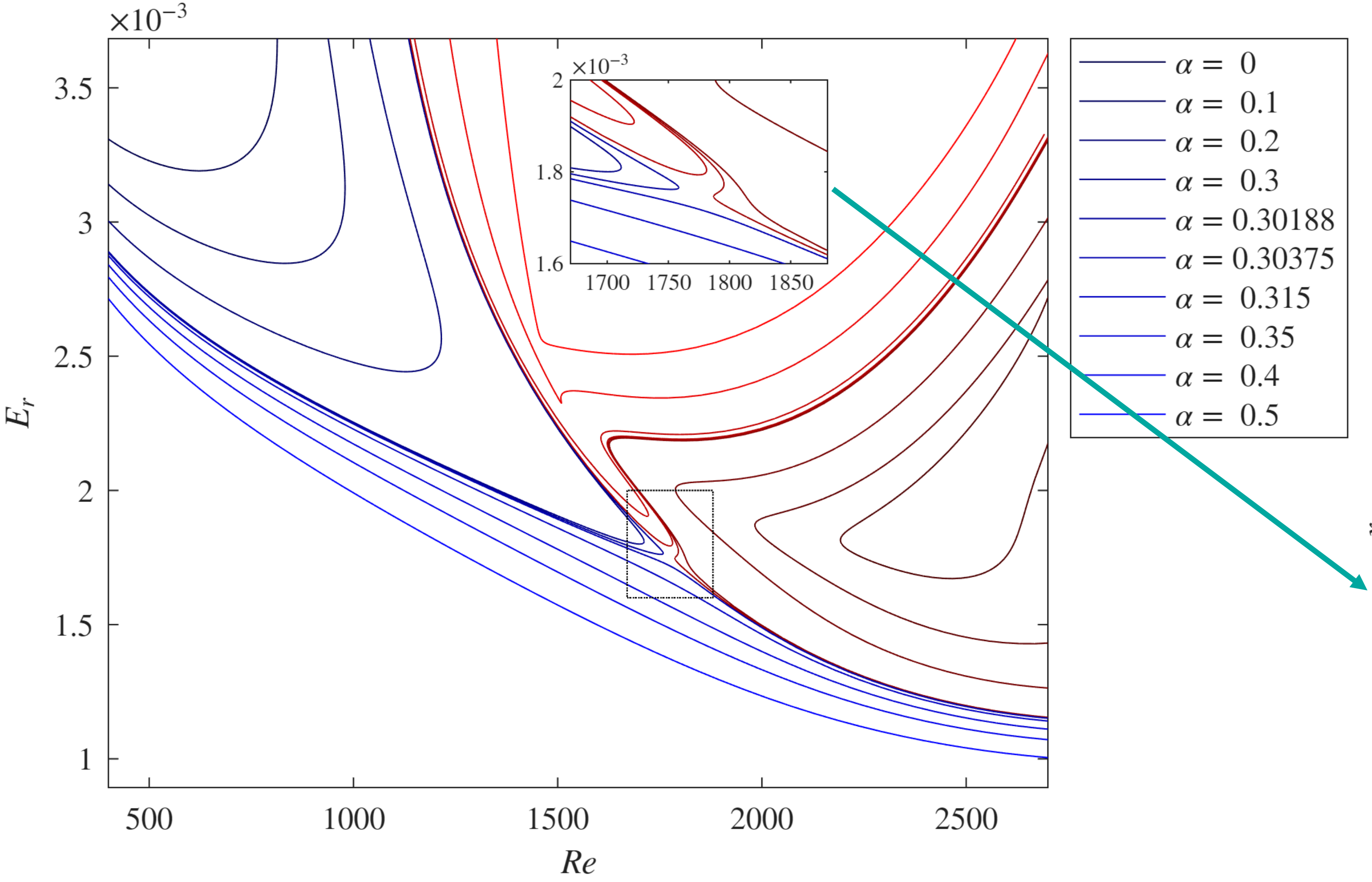
$$U_\theta \frac{d\Omega}{dr}\left[\frac{d\Omega}{dr}\frac{d\Gamma}{dr} + \left(\frac{dU_z}{dr}\right)^2\right] < 0$$

where $\Omega = U_\theta / r$ and $\Gamma = U_\theta\, r$.

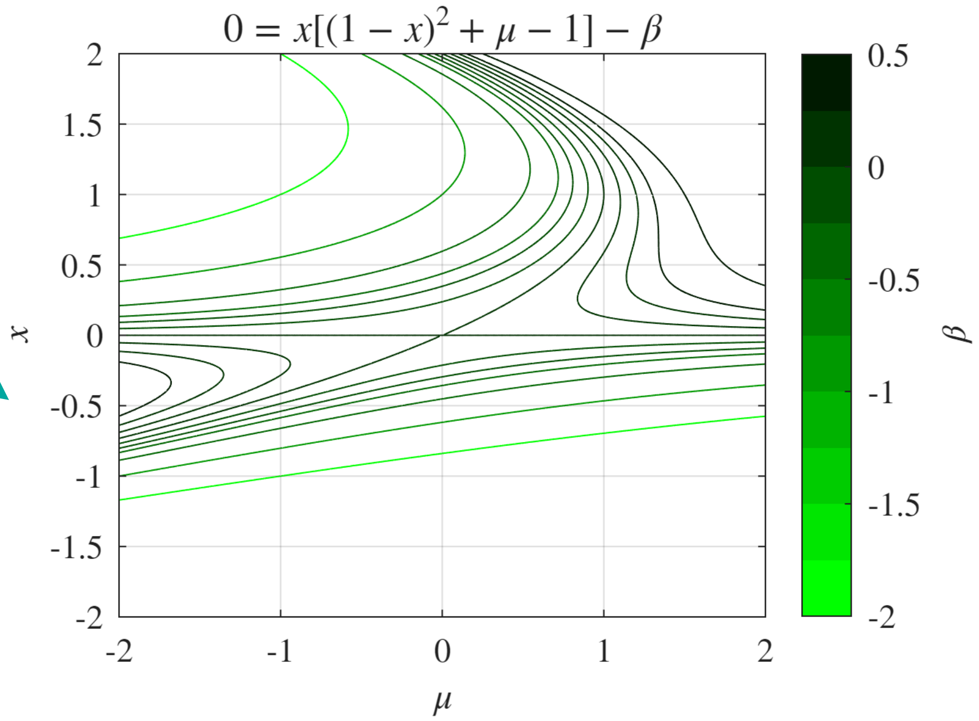



ψ [–]
1.3
1.1
0.9
0.7
0.5
0.14
0.16
0.18
0.20
0.22
0.24
0.26
0.28
0.30
0.32
0.34
φ [–]
70%
80%
85%
90%
91%
91.5%
85%
80%
20°
25°
30°

Transcritical bifurcation

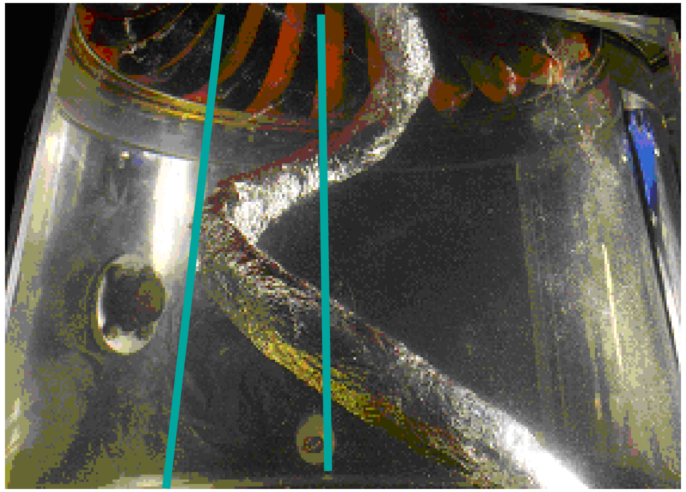


$$\dot{x} = x[(1-x)^2 + \mu - 1] - \beta$$

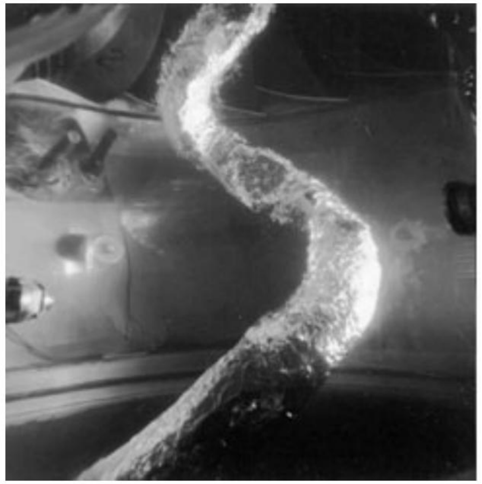


François Charru *Instabilités hydrodynamiques*

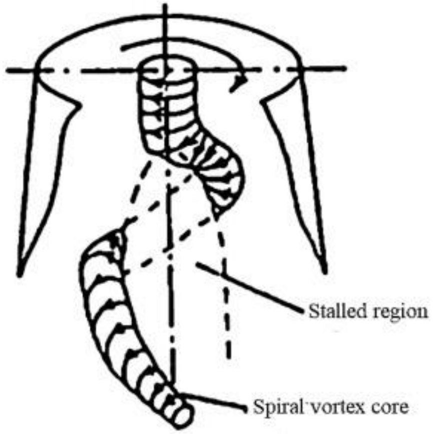
No-slip computations summary



Ciocian *et al.*, JFE 2007



Mauri, PhD 2002



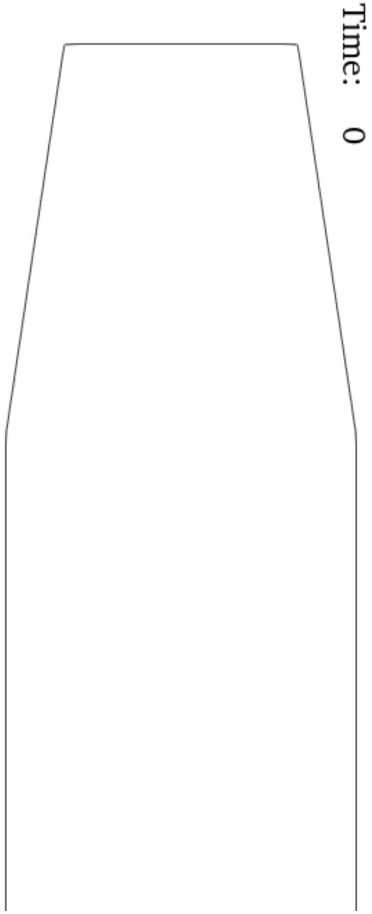
Nishi, IAHR Symp. 1982

Experiment (Re=10^6)

- Frequency $0.2\Omega - 0.4\Omega$
- Rope cone opening angle
- Stall region at the axis

Laminar no-slip wall simulation (Re ≈ 10^3)

- Vortex rope due to a supercritical Hopf bifurcation
- Rope frequency $\approx \Omega$
- No rope cone opening angle
- Vortex rope seems to be suppressed by the boundary layer
- Wall receptivity mode
- No clear stagnation region at the axis



Maybe we could change the wall boundary condition to free slip.